Biology - Summarised Notes

Infectious Disease: Module 7

Classifying Pathogens causing Disease:

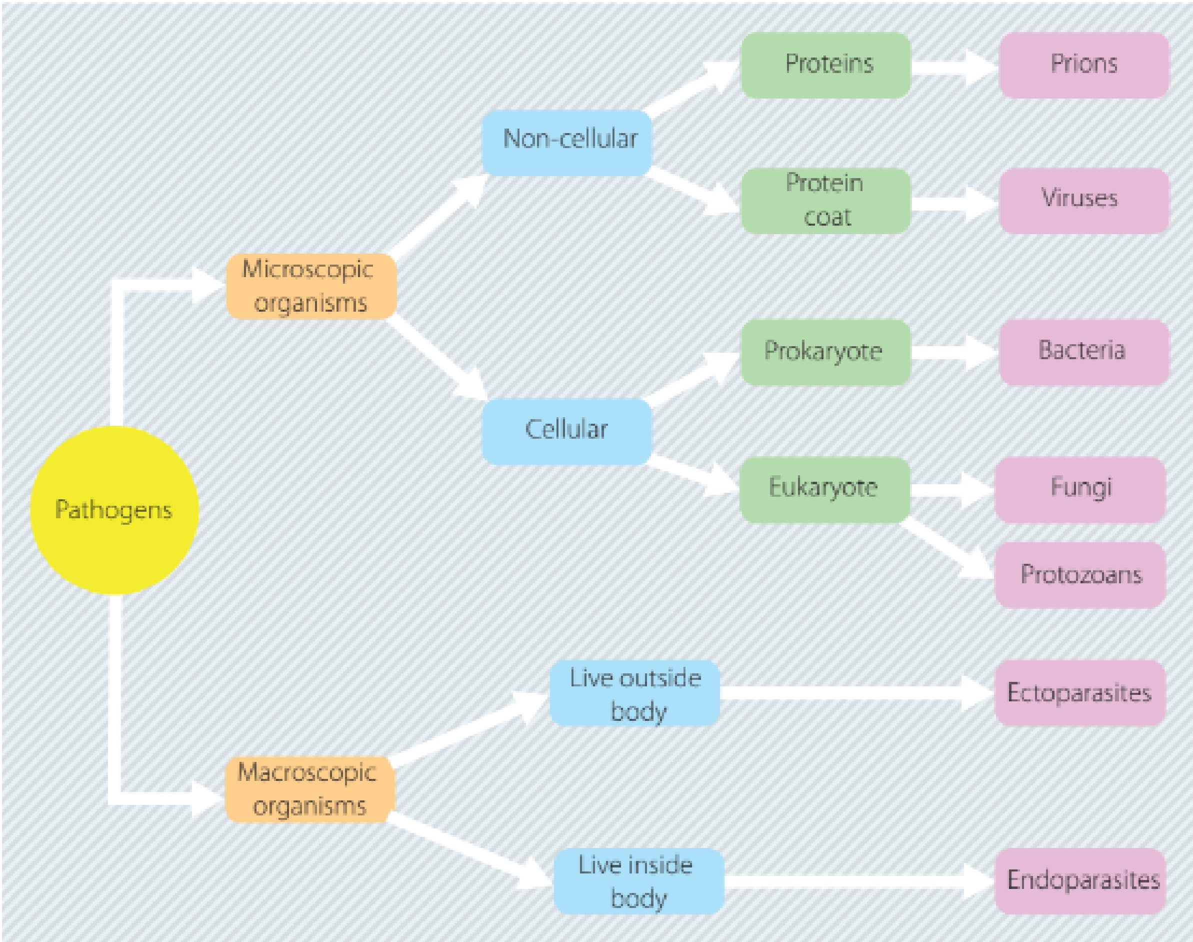
Infectious diseases:

- Disease:
    - Any condition that impairs the normal functioning of an organism
- Have causative agent (pathogen)
- Are usually contagious; can spread between individuals
- Examples:
    - Common cold (virus)
    - Tetanus (bacteria)
    - Hair lice (parasite)
- Communicable disease: diseases that plant → plant, animal → animal
- Virulence factors: Molecules which
    - Help pathogen entry into cells
    - Help avoid pathogen destruction by antibiotics
- Signs: physical factors
- Symptoms: factors reported by patient
- Pathogenic burden: pathogenic numbers and effect on the immune system
    - Increases in burden
        - associated with infectious disease since
            - body's immune defences become overwhelmed
- Likelihood of organism developing infectious disease depends on:
    - Pathogenicity (ability of pathogen to cause disease) of microbe, including numbers (burden)
    - Defence capabilities of host
- Loss of balance between defence and attack against pathogen = disease

Pathogens:

- Are causative agents for infectious diseases

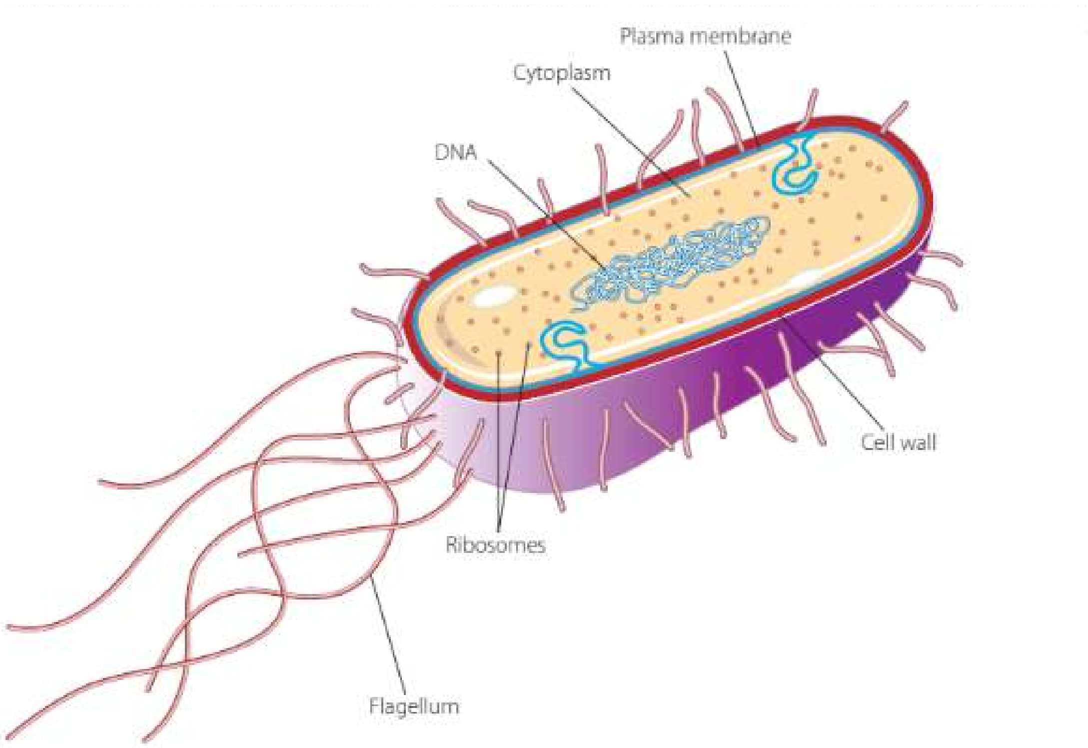
Classifying Pathogens:



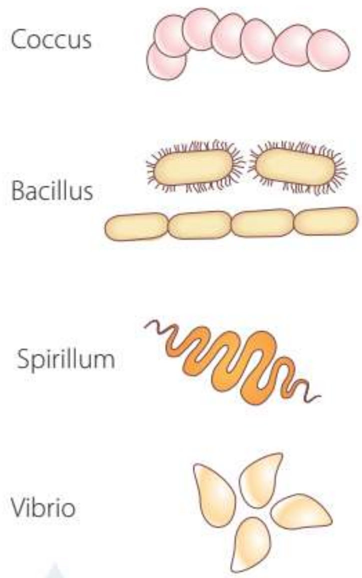
Bacteria:

Classification:

- Prokaryotic:
  - No nucleus
  - No membrane bound organelles
- Unicellular
- Cell wall
- Classed as 1 domain out of 3 (eukaryotes, bacteria, archaea)

- 

- Reproduce asexually
  - Binary fission: split into 2
  - Generation Time: Amount of time taken for bacteria to double
- 0.2 to 10 µm (micrometres) in length
- Genetic material either:
  - Bacterial DNA in large circular chromosome
  - Plasmids: Smaller DNA in fragments
- Cell wall; made of peptidoglycan (made of protein and carbohydrate molecules)
- Genus Mycoplasma - lack cell wall
  - Obligate intracellular parasites - can't live, reproduce outside cell
- Some bacteria have polysaccharide capsule - acts as virulence factor, being more effective in causing disease
- Classified by shape:
  - Spherical - coccus
  - Rod-shaped - bacillus
  - Spiral - spirillum
  - Comma shaped - vibrio



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- Different types of bacteria characterised by their staining properties:
  - Gram-positive
  - Gram-negative
- Respiration methods:
  - Aerobes (oxygenated environment)
  - Anaerobes (deoxygenated environment)
  - Facultative anaerobes (if oxygen - aerobic, if no oxygen - anaerobic; can switch between 2)

Transmission:

- Bacteria cause disease by
  - Producing toxins or chemicals which are harmful to hosts body
  - Damaging host tissue directly
- Parasitic relationship: bacteria benefits, host harmed
- Transmission may occur through
  - Direct - Close contact with infected organism
  - Indirect - touching bacterial contaminated object
- Treatment
  - Antibiotics
  - Concurrent management
    - Surgical removal of dead tissue
    - Wound cleansing
- Bacteria may form endospores (tough, waterproof external layer)
  - Lying dormant (not moving, inactive) in environment for years
  - Example: clostridium tetani
  - Endospores resist

        - Heat
        - Chemicals
        - Desiccation (drying out)
    - Normal methods of sterilisation ineffective
- Bacteria may aggregate
    - Biofilm - mucus like structure
        - Bind to living tissue
        - Carbohydrate matrix formed in it may bind bacteria to other surfaces like implants, catheters
            - Defends against antibiotics
            - Makes detection, elimination difficult

Examples:

Bacteria:	Name of Disease:	How disease caused and affects:	Features of disease:
Bordetella Pertussis	Whooping cough (pertussis)	- Affects lungs, airways - Transmitted via droplets through coughing, sneezing - Bacteria colonises and multiplies on mucous membranes of respiratory tract	- Runny nose - Sneezing - ‘Whoop’ cough during coughing bouts
Salmonella Enterica	Salmonellosis	- If infected, found in intestinal tracts of humans, animals, birds - Transmitted by consumption of foods contaminated by animal faeces	- Vomiting - Diarrhoea - Dehydration - Fever - Abdominal cramps

		- Bacteria invades and destroys cells lining intestines - limiting water absorption	
Mycobacterium Tuberculosis	Tuberculosis	- Bacteria settles in lungs, beginning to grow - Moving through blood from lungs to kidney, spine, brain - Transmitted through droplets by coughing, sneezing, etc.	- Cough - Fever - Night sweats - Blood-stained sputum
Clostridium Tetani	Tetanus (“lockjaw”)	- Infection usually occurs through contamination of wounds - Caused by spores of bacteria - Spores release toxin that is carried through blood cells into neurons, not being able to be neutralised	- Fever - Sweating - Headache - difficulty swallowing - rapid heartbeat - Muscle spasms starting with jaw, spreading to rest of body
Chlamydia Trachomatis	Chlamydial disease (sexually transmitted)	- May infect urethra, cervix, anus, throat, eyes - Spread through vaginal, oral, anal sex	- Burning sensation while urinating - Discharge - May lead to infertility if untreated

		- Invades mucosal epithelial cells of body parts by attaching themselves to the outside and getting engulfed as a vesicle by cell	

Fungi:

- Ideal conditions
    - Warm, moist environment

Classification:

- Chitin
    - cell wall
- Heterotrophic
    - need external food sources
- live on dead plant, animal material
    - Decomposers
- Unicellular or multicellular
- eukaryotic
- Composed of system of
    - microscopic tubular filaments or threads
        - hyphae
- Hyphae branch to from mycelium
- Microscopic or macroscopic
- Asexual reproduction (yeasts)
- Sexual and asexual reproduction

Transmission:

- Fungal infections (mycoses) may be:
    - Cutaneous: affecting outer skin layer
    - Subcutaneous: affecting under skin surface from penetrating wounds
    - Systemic: affecting internal organs
- Are opportunistic, affecting

- Weakened immune systems
- Concurrent diseases (diseases existing at the same time)
- Dermatophytes - live on skin, nails, hair
- Diseases transferred through
  - Close contact with diseased person, animal
  - Contaminated objects

Examples:

Fungus:	Name of Disease:	How disease caused and affects:	Features of disease:
- Epidermophyton - Trychophyton - Microsporum	- E - Tinea - T - Tinea - M - Ringworm	- Affects skin - Transmitted through direct contact, damp surfaces - Produce enzymes that break down keratin (a protein) in skin	- Cutaneous mycoses - Fungus secretes enzymes which break down keratin - Needs exposure to pathogen and warm, moist conditions - Red, itchy, scaly skin - Ringworm - ring shaped lesions (area of abnormal, damaged, tissue) on skin - Yellowing. Hardening of nails
Sporothrix schenckii	Sporotrichosis (rose gardener's	- Affects under the skin - Less commonly	- Subcutaneous mycoses - Fungal entry

	disease)	affects lungs, eyes
		- Very rarely affects other parts of body
		- Transmitted through
		  - Scratches from thorns
		  - Handling infected plant matter
		  - Scratches, bites from cats
			through small cut, scrape
			- Transmitted through
			  - Scratches from thorns
			  - Handling infected plant matter
			  - Scratches, bites from cats

- Histoplasma capsulatum - Blastomyces Dermatidis - Coccidioides spp.	- H - Histoplasmosis - B - Blastomycosis - C - Coccidioidomycosis	- Caused by spores of fungus - Affects lungs - Histoplasma - enters white blood cells, changing from a mould into yeast (infectious form) - Coccidioides - acute and chronic inflammation	- Systemic mycoses - Severe lung disease - Histoplasma grows in soil contaminated with bat droppings - Blastomycosis contracted by breathing fungal spores from soil, leaf material

Candida Albicans	- oral candidiasis
	- (thrush)
		- Overgrowth of yeast fungus
		  - Candida albicans
		- Yeast normally in less amount in body
		- Contributing
			- Affects
			  - Mostly mouth
			- May affect
			  - Skin of gential area around vagina
			  - In groin area in males

		factors - Antibiotic use - Weakened immune system - Hormonal changes - Poor dental hygiene	- Symptoms: - White, creamy patches in mouth, tongue - Redness - Cottony feeling in mouth - Vaginal thrush: - Itching - Burning - Thick white discharge

Protozoa:

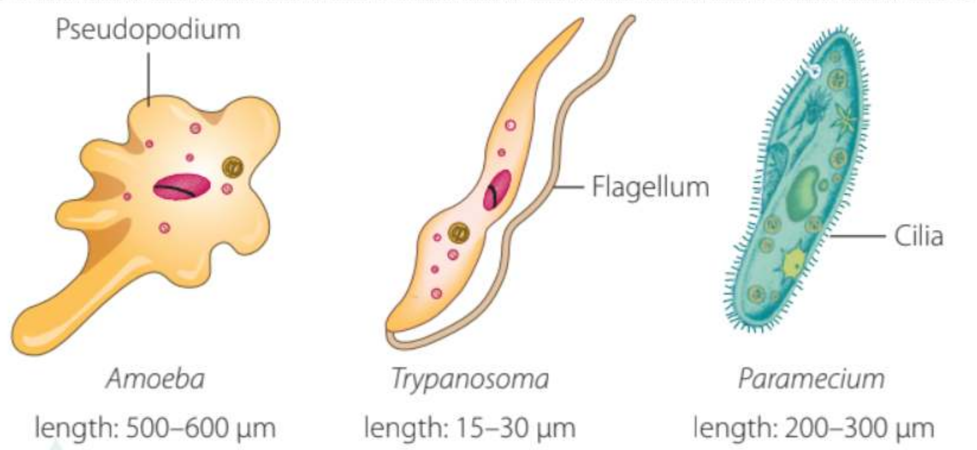
Transmission:

- Transmitted by
  - Insect bites
  - Contaminated water (faeco-oral route)

Classification:

- Unicellular
- Eukaryotic
- No cell wall
- Reproduce by binary fission
- Size - 1 - 300 µm
- Motile - able to move
- Flagellates
  - Move using flagellum (long, whip like tail)
- Ciliates
  - Move using cilia (hair-like projections)
  - Cilia beats rapidly
- Amoebae
  - Move using pseudopods (projection of cytoplasm)
- Sporozoa

- No structures of motion
- Reproduce by releasing spores



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- Protozoal diseases treated using antiprotozoal medications

Examples:

Protozoan:	Name of Disease:	How disease caused and affects:	Features of disease:
Plasmodium spp. - Plasmodium falciparum	Malaria	- Transmitted through bites of infected mosquitoes - Infect red blood cells - Protozoan enters red blood cells and multiplies - Red blood cells rupture, releasing protozoa and pathogen infects more red blood cells	- Fever - Fatigue - Headaches - Jaundice - Vomiting

Toxoplasma gondii	Toxoplasmosis	- Contracted by eating undercooked meat - From cat faeces - Reproduces in intestinal tracts of cats - Remain stuck inside cysts in body	- May be asymptomatic - Headache - Fatigue - Fever - Muscle aches - Tender lymph nodes - Seizures
Entamoeba histolytica	Amoebiasis	- Infects bowel - An infective cyst or trophozoite - Spread through human faeces, contaminated water - Kill cells	- May be asymptomatic - Diarrhoea (bloody) - Fever - Chills

Viruses:

- Genetic material in form of nucleic acids
- Able to pass on hereditary information upon entering host cell
- Non-cellular (no cells)
- Not free-living; only reproduce, metabolise in host cell

Classification:

- Protective protein coat (capsid) encloses genetic material (DNA or RNA)
- Retroviruses: viruses with RNA
- How virus works:
    - Virus attaches to surface of host cell with chemical on protein coat
    - Takes over host cells reproductive mechanisms
    - Cell makes many copies of virus
    - Cell bursts and dies with so many copies of virus
    - Viruses released take over other cells
    - Virus may use cell membrane of host cell to make own lipids, glycoproteins
    - Bacteriophages (virus for bacteria) inject genetic material into cell

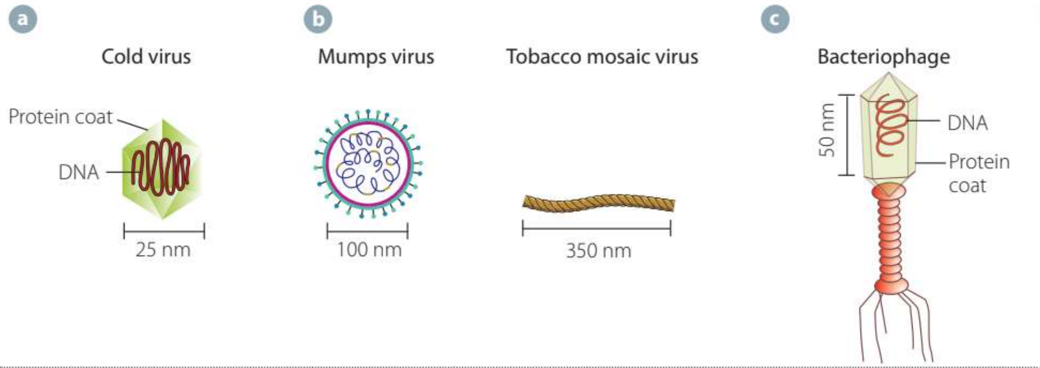
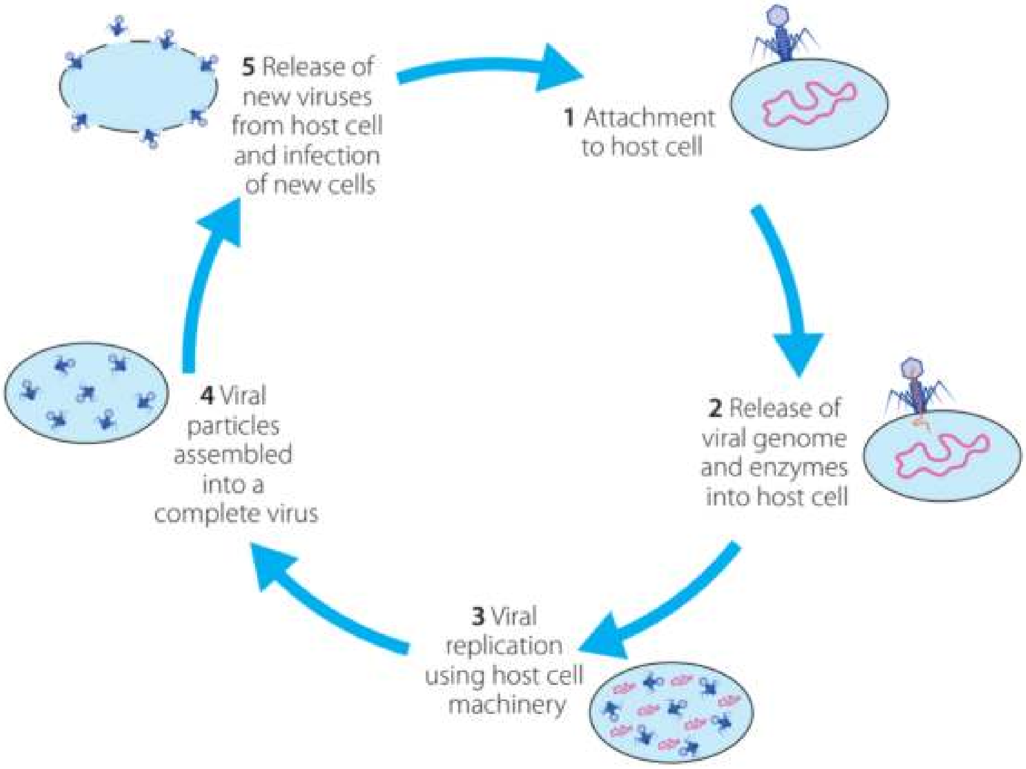


FIGURE 10.28 Structures of different viruses:
a icosahedral;
b spherical virion
c helical (filamentous);
d complex



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Examples:

Virus:	Name of Disease:	How disease caused and affects:	Features of disease:
Influenza virus	Influenza	- Direct contact through close contact or inhalation of infected droplets - Indirect contact touching	- Fever - Head, body aches - Coughing - Stuffy, runny nose

		contaminated objects - Infects respiratory epithelial cells - Effects synthesis of cellular proteins	
Human Immunodeficiency Virus (HIV)	AIDS	- Direct contact: sexual contact - Indirect contact: use of shared needles, shooting of illicit drugs, contact with infected blood - Kills white blood cells by invading them and replicating	- Fever - Muscle pains - Headaches - Sore throat - Night sweats - Mouth sores - Yeast infection (thrush) - Swollen lymph gland - Diarrhoea
Morbillivirus	Measles	- Airborne - transmitted through infected droplets in air - Increases susceptibility to long term infections - changes pathogen-specific antibody responses and types of cells in	- High fever - Cough - Runny nose - Rash all over body

		circulation	

Prions:

Classification:

- Abnormal proteins
- Non cellular
- Cause diseases of nervous system
- No genetic material
- Cause disease by inducing abnormal folding patterns of normal prions they come into contact with
- Abnormal proteins sent to central nervous system, other organs

Transmission:

- Can be contracted by:
    - Ingesting tissue with infectious prions
    - Getting contaminated growth injections by infected animals
    - Getting corneal transplants from organ donors (previously infected)
    - Inheriting mutated gene for infectious prions
    - Spontaneous creation of prions

Examples:

- Diseases:
    - Creudtzfeldt-jakob disease (CJD):
        - Rapidly progressive
        - Fatal brain disease
    - Variant CJD (mad cow disease) in humans from cattle
    - Cervidae:
        - Chronic wasting disease
        - Of north american deer, elk, moose
    - Scrapie:
        - From sheep, goats
    - Kuru:
        - Humans
    - Fatal familial insomnia:

        - Spontaneous formation of protein in brains of humans

Macroscopic Parasites:

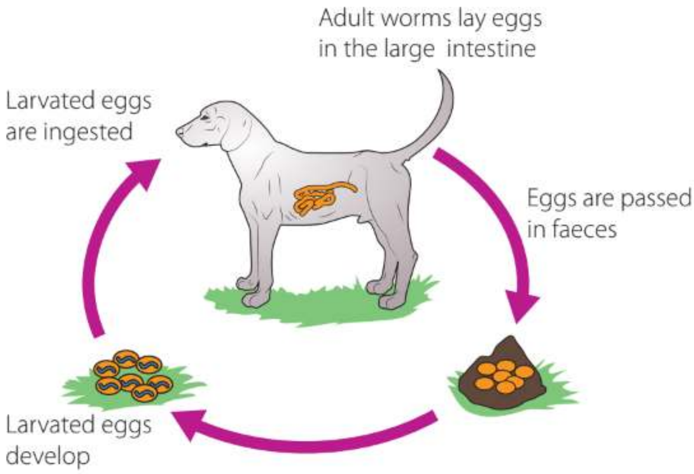
Classification:

- Multicellular
- Eukaryotic
- Can be seen with naked eye
- Endoparasites:
    - Live inside hosts body
    - Examples
        - Flatworms
        - Roundworms
- Ectoparasites:
    - Live outside of hosts body
    - Usually suck blood
    - Examples
        - Mosquitoes
        - Fleas
        - Ricks
        - Leeches
        - Mites
        - Lice
    - May inject toxins while feeding
    - Vectors of transmission of other pathogens

Helminths:

- Classification:
    - Worm-like
    - Inhabit gastrointestinal systems of humans and animals
    - Live on nutrients supplied by host
    - Disrupt digestion, absorption of nutrients
    - Affect immune response
- Transmission:
    - Eggs in environment
    - Eggs picked up by host through unsanitary drinking water, infected soil
    - Larvae from eggs and mature in host
    - Mature worms lay eggs in host

- Eggs passed through faeces of host



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Arthropods:

- Invertebrates
- Exoskeleton
- Segmented body
- Include
    - Insects
    - Arachnids
- Are ectoparasites
- May act as vectors for other pathogens
- Contribute to
- Blood loss
- Concurrent infections
- Examples
    - Fleas:
        - Humans, animals
    - Ticks:
        - Humans, animals
        - Inject neurotoxin
    - Lice:
        - Humans, animals
    - Mites:
        - Mostly animals also humans
    - Flies:
        - Transmit pathogens
    - Mosquitoes:

        - Biological parasites
        - Vectors, transmitting other pathogens

Non-Infectious diseases:

- Caused by:
    - Genetics
    - Environmental influences
    - Cellular malfunction
- Are not spread between individuals
- Examples:
    - Cystic fibrosis (genetic)
    - Down syndrome (genetic)
    - Sunburn (environmental - excess sun exposure)
    - Cancers (uncontrolled cell division - mitosis)
- Multiple factors may contribute development of non-infectious disease
    - Example:
        - Genetic disposition + cellular malfunctions
            - BRCA1 gene for breast cancer
        - Genetic disposition + poor diet
            - Type 2 diabetes

Transmission during Epidemics

- Epidemic:
    - Increase in the no. of people affected by a disease
    - Occur when increase in no. of people affected is more than normal
- Is particular disease in particular area
    - Area: several communities
- Endemic:
    - More people affected by disease than normally
- Outbreak:
    - Disease happening temporarily in isolated area (a school, a university)
- Pandemic:
    - Increase in no. of cases of disease across a continent or internationally

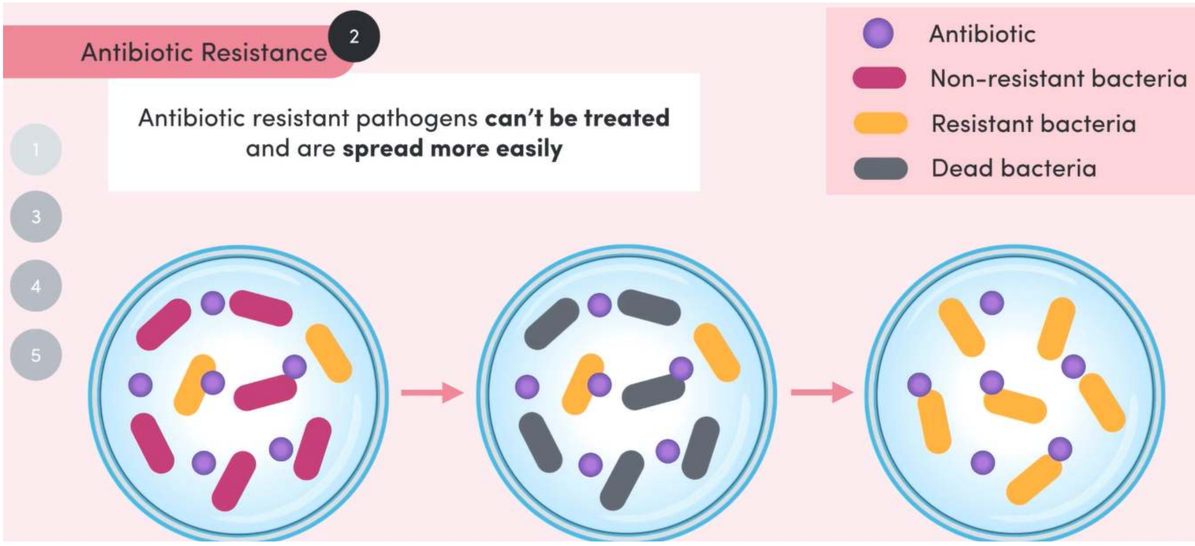
Factors affecting Epidemics:

Pathogen:

Virulence:

- Pathogen's ability to infect, damage host
  - More virulence → more effect on health of host
  - Mutations increase pathogens virulence
    - Evade detection in hosts

Antibiotic Resistance:

- 

Toxins:

- Make hosts more susceptible to infection
- Deter other pathogens

Genetic Shift:

- When there is loss in genetic variation, selection pressure that causes population to lose genetic resistance

Herd Immunity:

- Group where most individuals immune → population becomes immune
- Pathogen can't easily spread → deterred epidemics

Human:

Migration:

- People - disease vectors
- Increased movement of people → more chance disease spread between them

Infrastructure:

- When populations are dense, close together
  - → pathogen exposure more likely, increased
- Disease survival by lack of
  - Clean, running water
  - Effective sewage systems
  - Adequate healthcare

Healthcare:

- Disease spread increased when no access to drugs, vaccines
- Can be because of
  - Lack of access to infrastructure (clinics, pharmacies)
  - High cost of medicine

Controlling Epidemics:

Identifying Pathogen:

- Health workers identify disease through:
  - Clinical observation
  - Laboratory observation
  - Data collected on cases shown and put into large surveillance systems

Environmental Management:

- Change environment to reduce contact with pathogen
- Includes:
  - Cleaning water supplies from disease
    - Examples:
      - Boiling
      - Chlorinating
      - Sealing
  - Reducing food contamination risk
    - Examples:
      - Limiting food prep
      - Disposing of old food
  - Creation of sanitary conditions:
    - Examples:
      - Removing waste from public areas
      - Providing sealed sewage systems

Quarantine:

- Period of restricted movement of and separation of people, animals, materials which may spread infectious disease
- Large-scale quarantines:
    - Travel bans
    - Border regulation
- Small-scale quarantines:
    - Protective clothing
    - Isolation

Microbial Testing of Food and Water samples:

Aim:

- To compare fresh and old food by the prevalence and type of microbial colonies present.

Hypothesis:

- The agar plates holding old food will produce a greater quantity and variety of microbial colonies since older food would have been exposed to microorganisms for a longer period of time than with freshly made food.

Variables:

Independent:

- Age of food

Dependent:

- Number and type of microbial colonies present on agar plates

Investigation control:

- An agar plate with no food
- Used to verify that the microbes on plates are from food samples and are not present in the agar to begin with.
    - Validates experiment by confirming that microbes come directly from the food

Controlled (Constant):

- Type of agar
- Incubation time
- Incubation temperature

- Type of food used

Risk Assessment:

Identification of Risk:	Potential Harm:	Assess:	Control:
Agar plates with microbe colonies	- Exposure to colonies initiating disease	- Moderate	- Seal off agar plates after exposure - never open again - Sterilise after se and dispose of correctly
Sterilising inoculating loop	- Burns from hot wire after heating	- Moderate	- Only touch handle of inoculating loop - Practice caution around object
Exposure to microorganisms	- May cause disease	- Low	- Wipe benches with disinfectant - Use disposable gloves - Wash hands with soap after contact with plates

Equipment:

- Agar plates
- Old food
- Fresh food
- Inoculating loop

- Bunsen burner

Method:

1. Swab bench with disinfectant solution
2. Collect inoculating loop (wand)
3. Practice various aseptic techniques that your teacher shows you.

   This includes:

   - correct sterilisation of your inoculating loop
   - correct streaking procedure
   - the need for re-sterilisation of the loop in between streaks
   - opening agar plates for minimum time necessary and at a small angle
4. Streak plates relative to their need (foods old vs fresh) + control group
5. Seal and label all plates, leaving a 1 cm gap with no tape to allow condensation to escape
6. Incubate plates at 37 degrees celsius for 3 days
7. Remove plates from incubator and record results
8. Repeat steps 1-7 at least 3 times for 3 trials

Results:

<table>
<tr><th>Bacterial Colonies:</th><th>Fungal Colonies:</th></tr>
<tr><td>- Shiny
- Smooth
- Glossy
- Coloured
- Round uniform shape
- </td><td>- Fuzzy
- Fluffy
- Powdery
- Splat like shape
- Large
- </td></tr>
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Modes of Transmission:

- Needed for disease to spread:

    - A host susceptible to disease
    - Pathogen able to carry disease
    - Mode of transmission - a way for pathogen to go from host to host

Direct Contact:

- Transmission by physical contact between a host and non infected organism
- Horizontal Transmission:
    - Contact between organisms of the same generation
    - Not parent and child
- Vertical Transmission:
    - Contact between parents and their offspring
    - Example:
        - Mother to baby during childbirth
- Physical contact includes:
    - Touching
    - Sexual contact
    - Kissing
    - Contact with nasal, oral secretions
    - Biting
    - Direct contact with body fluids (blood, semen)
    - Direct contact with wounds
    - Prenatal (before birth, during pregnancy)
    - Perinatal (around time of birth)
- Examples of Diseases:
    - Skin infections (ringworm, impetigo)
    - Cytomegalovirus (CMV)
    - Glandular fever
    - aids
    - Herpes simplex virus

Indirect Contact:

- Transmission by no direct contact between host and another organism
- Infection happens through an area made by host outside of itself
- Fomite:
    - Object, substance carrying infection, pathogens
- Indirect means of transmission:
    - Airborne - coughing, sneezing

    - Touching infected surface
    - Contaminated food, water
    - Infected surgical instruments not sterilised correctly
    - Vectors - mosquitoes, ticks, fleas
- Examples of Diseases:
    - Measles - infected droplets
    - Gastroenteritis - contaminated food water
    - Toxoplasmosis - infect cat droppings
    - Legionnaires disease - contaminated water in cooling towers
    - Influenza - droplets, biological matter

Vehicle Transmission:

- Spread of pathogens by contaminated air, food, water

Vector Transmission:

- Animals or organisms assisting in transfer of pathogens between host to uninfected individuals or organisms

Biological Vectors:

- Pathogen undergoes part of its life cycle in vector
- Example:
    - Mosquitoes - malaria

Mechanical Vectors:

- Do not become infected with pathogen themselves

Koch's Postulates:

- Robert Koch discovered:
    - Pathogens for anthrax, tuberculosis, cholera
- He created set of principles to identify disease: Koch's Postulates
- Developed agar plate technique for culturing microorganisms

Postulate Principles and Method:

- Principles to determine if microorganism is responsible for a disease

1. Same microorganism must be present in every host
2. Microorganism must be isolated and cultured in laboratory
3. When sample of pure culture in put into healthy host, host must develop same symptoms as original host

4. Microorganism from second host must be isolated and cultured, being identified as the same microorganism from original host

Cause and Transmission of Anthrax:

- Got infected blood of dead sheep with anthrax
- Observed blood with microscope to find:
    - Active rod-shaped cells
    - Inactive dormant spores
- Isolated rod bacteria; growing them in cultures
- Injected cultured bacteria into other sheep and mice
- Sheep and mice developed anthrax
- Showed that cause of anthrax was anthrax bacteria (microorganism) - spontaneous generation ❌
- Showed that bacteria grown outside organism caused disease - germ theory ✅
- After Koch's study:
- Determined transmission of anthrax:
    - Animal not contacting diseased animals still got anthrax
    - Dead bodies of infected animals were buried in grazing fields
    - Spores of anthrax bacteria from dead bodies of animals were there

Pasteur's Experiments:

- Found microbes responsible for fermentation or creation of alcohol in beer, wine
- Found microbes responsible for souring of alcohol, spoiling of milk
- Found that microorganisms were in open air - spontaneous generation ❌
- Found vaccines for chicken cholera, anthrax, rabies

Pasteurisation and Fermentation:

- Sampled fermenting wines under microscope
- Found yeasts - converting sugars → alcohol
    - Found process of Fermentation
- Found bacteria - converting sugars → lactic acid
- Bacteria found in wine, beer, milk caused spoilage
- Created process of Pasteurisation:
    - Showed that heating wine, beer for a period of time after fermentation killed remaining bacteria
    - Heating milk killed bacteria in it

Swan-Necked Flask:

- Meat broth boiled in swan necked flasks
- Air drawn in through flasks when broth cooled
- No bacterial growth
  - Microorganisms in air got trapped in curved neck of flask
  - No bacterial growth happened in flasks
- Bacterial growth
  - When neck of flask broken off - bacterial growth occurred
  - When broth tipped to curve of flask where microorganisms trapped - bacterial growth occurred
- Experiment showed that microorganisms contaminating broth were carried in air not being spontaneously generated; germ theory ✅, spontaneous generation ❌

Effects of Disease on Agriculture:

- Agriculture:
  - Involves cultivation of crops, pastures, rearing of animals to provide for humans:
    - Meat
    - Milk
    - Fibres
    - Other products
- Endemic diseases: Diseases constantly present within a country, region
  - Examples:
    - Bovine Johne's disease - cattle, sheep, goats
    - Anthrax - sheep, cattle
    - Footrot - sheep
- Exotic diseases: Diseases that have been introduced to Australia
  - Examples:
    - Foot and mouth disease
    - Avian influenza - birds
    - Bovine tuberculosis
    - Equine influenza
    - Newcastle disease - domestic poultry, wild birds

Factors - Development of infectious disease:

- Host factors:
  - Susceptibility to disease
  - Access to pathogen

        - Concurrent disease, infection → weakened immune response
        - Drought, heatwave stress on host
    - Pathogen Factors:
        - Pathogens availability
        - Ability to transfer between hosts
        - Virulence factors:
            - Adhesion to and invasion of host
            - Living and starting disease inside host tissues
    - Environmental Factors:
        - Overcrowding
        - Lack of hygiene → buildup of wastes
            - Good environment for pathogen reservoirs
            - Good environment for pathogens to establish, cause disease

Factors - Increased risk of infectious disease:

- Increased movement - human population:
    - Travellers, imported livestock, plants - carry disease into Australia
    - Pathogens form reservoir in:
        - Food
        - Soil
        - Seeds on shoes
        - Infected animals
        - Animal products
    - ↑ reservoirs moved through different areas by humans
- Increase - intensive, industrial-type agriculture:
    - Increase in human population → use of intensive feedlots
    - Feedlots - animals in high stocking densities
        - Increasing rate of pathogen transmission
    - Animals exposed to own waste products - bacteria
- Changing patterns of land use:
    - Deforestation, irrigation - change distribution of insects
    - Habitat loss → bats closer to humans, horses (hendra, nipah viruses)
- Climate Change:
    - Insect vectors - changed distribution, abundance
    - Changes in rainfall patterns → may favour formation of reservoirs in soil, plants, insects
    - Ecosystem changes → affect availability of nutrients for animals, plants - reducing pathogen immune responses
- Antimicrobial resistance:

- Antimicrobials - treat infections in livestock
- "Off-label use": use of antibiotics in unapproved way
- "Off-label use" or overuse of antibiotics allows faster development of antimicrobial resistance ← natural selection of resistant bacteria
- Resistant bacteria transferred to humans by:
    - Direct contact with animals
    - Consumption of their meat
    - Transfer of genes - animal pathogens → human pathogens:
        - Normal bacterial infections - not able to be treated
- Antimicrobials used in high density situations

- Pesticide resistance:
    - Resistant forms of macroparasites, weeds - overuse of:
        - Insecticides
        - Acaricides
        - Herbicides
        - Anthelmintics
- Loss of Genetic Diversity:
    - Reduced resilience of population to new pathogen by:
        - Use of inbreeding - animals, plants
        - Monoculture (one crop) practices
- Increase - "Hobby Farmers":
    - People who farm or do animal husbandry for fun
    - Have little knowledge, experience → unaware of risks of certain practices
- Increase - Use of aquaculture as marine, freshwater animal populations decrease:
    - Aquaculture: farming of seafood
    - Human population increase → need for protein source increase
    - Wild stocks decrease - overfishing, habitat loss, marine pollution
        - Species:
            - Fish (salmon, tuna, barramundi)
            - Oyster
            - Abalone
            - Crab
            - Prawns
            - Lobsters
    - Pressure to use antimicrobials off-label - no current registered for aquaculture

Animal Diseases:

Causes and Effects:

Causes:	Effects:

Same as "Factors contributing to risk of disease"	- Death of infected animals (anthrax, blacks disease)
	- Loss of appetite, weight over short or extended period
	- Economic loss for farmer:
	    - Low profit, production from reduced yields of:
	    Meat
	    - Milk
	    - Wool
	- Loss of international trading opportunities - threatened disease free status of Australia
	- Human illness, disease
	- Low growth rates in young animals
	- Embryonic death, stillbirths → Loss of fertility in females
	- Blemishes, ectoparasites → lost economic value of individuals (warts in beef cattle)

Lyme Disease - Cattle:

- Pathogen:
    - Bacteria
    - *Borrelia*
- Transmission:
    - Through Bites of ticks (vector)
    - Ticks transmitted through direct contact
- Dairy cattle may develop:
    - Fever
    - Stiffness
    - Swollen joints
    - Lameness (not able to walk correctly)
    - Decreased milk production

- Fall in quality, quantity of product output → decrease in demand for cattle

Plant Diseases:

Causes and Effects:

Causes:	Effects:

Pathogens:
- Fungi:
  - Fungal spores in reservoirs:
    - Contaminated seeds
    - Farm machinery
    - Soil
    - Nearby weeds
  - Transmitted by:
    - Wind
    - Water
    - Contact with reservoirs through farming operations
  - Damage plant by:
    - Destroying conducting tissues
    - Absorbing nutrients from plants
- Insects, Mites
  - Direct damage to plant tissue
  - Vectors for pathogens
- Bacteria:
  - Reservoirs in:
    - Soil
    - Weeds
    - Seeds
  - Humans having bacteria from previous contaminated plants
  - Conditions for multiplying and spread:
    - Humid, warm weather
    - Overcrowding plants
    - Bad soil conditions (water, nutrients, pH, salinity)
	Biological:
	- Death of plant:
	  - Diseased conductive tissue → losing water balance
	  - Loss of photosynthetic tissue → not able to make food
	  - Cell death by:
	    - Pathogen attachment
	    - Effects on it by diseased photosynthetic, conducting tissues
	- Abnormal growth:
	  - Interference with production, distribution of hormones
	  - Problem with chlorophyll production → leaves turn yellow
	  - Mosaic patterns - viral infections
	  - Plants losing more water than taken from root → wilting
	  - Wilting caused by:
	    - Root damage
	    - Interference with conducting tissues
	
	Social, Economic on farmer:
	  - Less yields of grains, pastures, fruits, vegetables
	  - Economic loss for farmer → financial hardship, stress for family, local community

- Poor air circulation'
- Nematodes:
  - Live in soil
  - Root Knot nematode - tomato:
    - Attacks plant roots - creating galls, lumps
    - Eggs in soil
- Viruses:
  - Obligate intracellular parasites
  - Stable in environment
  - Stay in plant material after cropping
  - Reservoir - contaminated equipment
  - Transmission:
    - High plant densities
    - Frequent handling of plants by humans
- Phytoplasmas:
  - No cell wall
  - Transmitted plant to plant by insect vectors
  - Live in phloem tissue
 	
Abiotic Factors:
- Temperature variation
- Light availability
- Chemical agents (natural, synthetic)
- Water quantity, quality
- Nutrient availability in soils
	Social, Economic on Australia's economy:
	- Loss of trading opportunities - national, international

Panama Disease - Bananas:

- Affected Cavendish bananas
- Affected property growing them in Tully Valley, Queensland, Australia
- Pathogen:
  - Fungus
  - *Fusarium Oxysporum*
- Causes:
  - Yellowing, wilting leaves

    - Splitting stems
    - Damage to conductive tissues - plant starved of water, food
- Transmission:
    - Root-to-root contact
    - Contaminated soil from machinery, shoes
- Affected property was sold to Australian banana growers council
- Pathogen contaminates soil permanently
- Remains biosecurity risk
- No commercial growing of plants on area
- Price of bananas was high during outbreak
- Threat to banana export industry - south, central america

Adaptations of Pathogens:

- For organisms to cause disease, they must:
    1. Enter the host
    2. Multiply in host tissues
    3. Resist or not stimulate host defence mechanisms
    4. Damage the host
- Adaptations in pathogens assists:
    - Entry (adhesion to and invasion) into host
    - Transmission of pathogen from host to host
- Adaptations form a part of the virulence factors of pathogen

Adhesion to and Invasion (Entry):

Bacteria:

Adhesion:

- Pilli
- Fimbria
- Adhesins - Specialised surface proteins:
    - Bind to receptors on cells
    - Resist urine, mucus, cilia washing actions
- Translocation of bacterial proteins - allows cell membrane to engulf bacteria
- Biofilm - resist immune responses
- Changing surface structures - evade immune system

Invasion:

- Enzymes, toxins:

    - Break down cell contents
    - Disrupt cell membranes
- Capsules resist phagocytosis
- Intracellular bacteria: phagocytosis by macrophages, walling off in granulomas
- Chemical strategies - destroy host immune defences
- Host cell cytoskeleton - intracellular movement of bacteria
- Toxins secreted (endo, exotoxins) - damage host cells

Fungi:

Adhesion:

- Cell wall, capsule molecules - adhesion to host cells

Invasion:

- Thermotolerance - heat shock proteins cope with body temperatures higher than air
- Saprophytic mycelium → parasitic yeast when exposed to heat
- Cell wall, capsules protect from host attacks
- Hydrolytic enzymes - damage host cell, provides nutrients to fungus
- Evasion:
    - Capsule production
    - Limiting cytokine production in cells
    - Reduced fungicidal power in macrophages
- Opportunistic infections in immunocompromised patients

Protozoa:

- Toxoplasma Gondii:
    - Microtubule protrusion - entry into cell
    - Vacuolar membrane - protection from lysosomes

Virus:

Adhesion:

- Viral surface proteins adhere to cell surface receptors + co-receptors

Invasion:

- Enter nucleus of host cell to allow replication of virus gene
- endocytosis (movement of virus into cell): Enveloped viruses (influenza) are enclosed in envelope (endosome) formed by host cell membrane when moving into cell

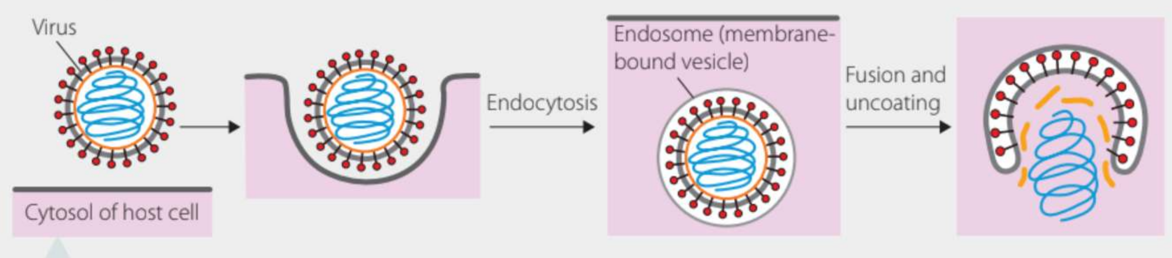

FIGURE 10.62 Receptor-mediated endocytosis of an influenza virus

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- Non-enveloped viruses form pore in host cell membrane and inject viral genome through it
- Some viruses use cells normal membrane forming process, travelling through endoplasmic reticulum and golgi body, then budding off from surface
- Frequently mutate surface proteins to avoid detection by hosts immune system

Prions:

- Host B lymphocytes: secrete factors that allow prions to invade dendritic cells in lymphoid tissue
- Lymphoid tissue → invade nervous tissue through autonomic nerves and travel to brain
- "Piggybacks" other proteins (ferritin - in meat) to facilitate movement through gut

Macroscopic Parasites:

Hookworms:

- Secrete immunomodulatory proteins - reduce host cell immune responses
- 3rd larval stage (L3) in soil invades host via hair follicles
    - Moves to lungs, trachea, intestine
- Teeth in buccal capsule anchor worm to gut lining

Ticks:

- Highly specialised mouthparts - attach to host skin
- Tick anchored by "attachment cement"
- Biologically active molecules - secreted to:
    - Prevent vasoconstriction (smallening of blood vessels)
    - Prevent host from forming clot
    - Prevent inflammatory response

Transmission:

Airborne:

- Spread through respiratory droplets in air
- Examples:

- Sneezing
- Coughing
- Talking

Adaptations:

- Suspended in air for long periods of time
- Resist drying out
- Initiates sneezing, coughing → encourages transfer to new host
- Aero-tolarant: tolerate wide range of oxygen concentrations

Examples:

- Influenza virus

Waterborne:

- Spread through pathogens in water

Adaptations:

- Colonise, multiply in water - environmental reservoir (storage area) like faeces present
- Have outer surface structures to allow movement (flagella, fimbria)
- Halotolarant: tolerate high salinity
- Not killed by boiling water or normal water treatment processes

Examples:

- Legionella
- Vibrio
- Giardia
- Campylobacter species

Vector-borne:

- Spread through living organisms

Adaptations:

- Changing type and number of vectors as storage area and transmission of pathogens
- Vector not affected by pathogen
- Pathogen uses host as reservoir in salivary glands or digestive tract for easier transmission
- Use surface proteins to attach to host
- Pathogen life cycle synced with feeding habits of hosts

Examples:

- Rickettsia felis - uses fleas, mosquitoes as vectors

- Malaria
- Zika virus
- Hendra virus
- Lyssa virus

Faeco-oral:

- From infected faeces → mouth of host
    - Faeces → flies → food → host
    - Faeces → fields/floors → food → host
    - Faeces → fingers → food → host

Adaptations:

- Stable in varied environments
    - Acid in stomach
    - Low oxygen
- Induce vomiting, diarrhoea - easing transmission
- Antimicrobial resistance genes

Examples:

- E. coli
- Salmonella species

Soil-borne:

- Pathogens spread + live in soil

Adaptations:

- Form endospores - resist drying out
- Survive in environment under conditions
- Grow in root zone (rhizosphere)

Examples:

- Clostridium tetani
- Fungi
- nematodes

Sexual (venereal):

- Spread through sexual activity, contact

Adaptations:

- Transmission across placenta - maternal, foetal cells next to each other

- Uterine invasion
- Transmission through unprotected sexual activity
- Eating of placenta by wild animals
- Pathogen particles may be suspended in air from afterbirth

Examples:

- Chlamydia
- hiv/aids
- Gonarrhoea
- herpes simplex virus

Blood-borne:

- Through blood

Adaptations:

- Take advantage of features of red blood cells for own growth, development

Examples:

- Sickle cell anaemia (inherited, non infectious)
- malaria

Vertical (mother to child):

Adaptation:

- Same as sexual

Examples:

- Brucella species
- Parvovirus
- Rubella virus
- Chickenpox virus
- Listeria monocytogenes
- Plasmodium falciparum - malaria

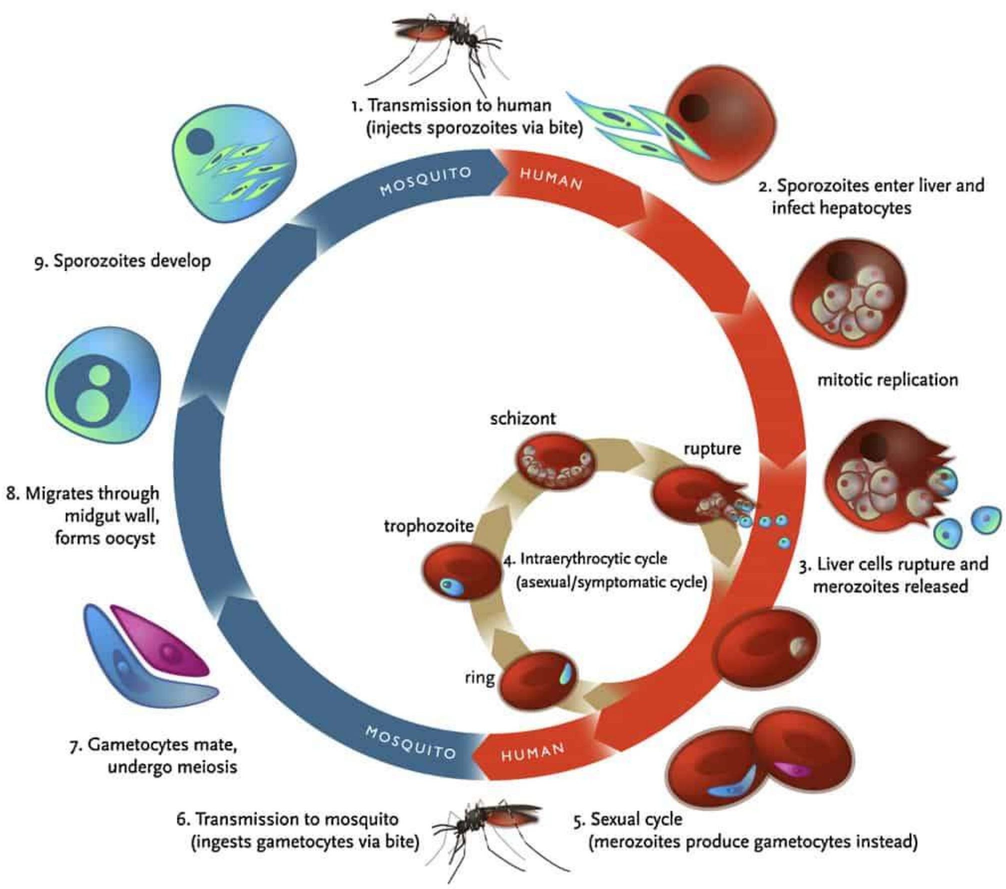
Case Study - Malaria

- Caused by genus plasmodium
- Protozoan
- Species:
    - Plasmodium falciparum (main)
    - Plasmodium vivax

    - Plasmodium ovale
    - Plasmodium malariae
- Plasmodium Falciparum - main cause of malaria and most dangerous

Life Cycle:

- https://www.youtube.com/watch?v=QGxn6MEmlnY
- Mosquito lands on human host for blood
- Injects saliva with sporozoite
- Sporozoites move to liver
- Divide rapidly in liver cells (schizogony)
- Sporozoites → merozoites
- Merozoites invade other liver cells
- Then invade bloodstream
- Invade erythrocytes in blood
- Enlarges into ring trophozoite
- Nucleus of trophozoite divide into schizont
- Schizont divides into merozoites
- Erythrocyte ruptures with many merozoites
    - Fever and chills
- Some merozoites develop into gametocytes - male and female
- Erythrocyte with gametocytes don't rupture
- When mosquito sucks blood of human, erythrocyte with gametocytes are taken into mosquito
- Gametes are formed in mosquito
- Gametes form diploid zygote in gut of mosquito
- Gametes → diploid zygote → oocysts → sporozoites
- Sporozoites move to salivary gland of mosquito to be injected into more hosts



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Adaptations of Plasmodium Falciparum:

Adhesion to and Invasion (Entry):

- Different life stages
- Each life stage, produces different antigens (molecules recognised as foreign)
    - Prevents host from carrying out effective immune response
- When in red blood cells, adhesion proteins produced on surface of cell
    - alters their movement through capillaries
    - Helps reproduction, survival of parasite
- Alter adhesion proteins on red blood cells
    - Protecting against immune system (3rd line of defence)
- Kill liver cells
    - Creates gap through which parasite move to infect other liver cells
- Collect calcium ions from other liver cells
    - Used to block appearance of antigen on surface of host liver cell
        - Avoiding immune system

Transmission:

- Female anopheles mosquito
    - Has anticoagulant protein (anophelin) - injected into human

        - prevents clotting
    - Plasmodium exits mosquito salivary gland after anticoagulant protein injected

Plant Responses to Pathogens

- Plants will be resistant to a pathogen if:
    - It prevents pathogen from invading plants tissue
    - It Prevents pathogen from reproducing
- Plants lack immune cells and adaptive immune system which is in mammals
- If pathogens enter:
    - Each plant cell works independently or
    - In unison with those directly associated with it to respond to invading pathogens
- Possess an innate immune system

Passive Defences:

- Are mechanisms that are present before contact with pathogen
- Mainly consist of physical barriers

Physical Barriers:

- Thick Cuticle:
    - Prevents entry of
        - Pathogens able to break down cuticles by secreting enzymes
- Bark:
    - Extra protection
        - from pathogens which may invade into food source, sap in phloem of tree
- Vertically hanging leaves:
    - Do not accumulate water film → reduced likelihood of pathogen reservoirs building up on outside surface of leaves
- Epidermis: skin of plant
- Stomata:
    - Can close when signalled
- Cell walls:
    - Made of lignin and cellulose

Chemical Barriers:

- Chemical compounds in plant tissues:
    - Reduce fungal, bacterial growth
    - Ward off vectors of viruses

        - E.g:
            - Glucosides
            - Saponins
- May produce enzymes to break down toxins from pathogens
- Chemical receptors detect presence of pathogen:
    - Detecting Pathogen Associated molecular patterns (PAMPs) - secreted by bacteria

Active Defences:

Gene-for-Gene Resistance:

- Involves specific interaction with plant + pathogen
- Plant has resistance genes that encode R proteins which recognise specific pathogen effects
- R proteins detect presence of pathogen Avirulence (AVR) genes using their protein products
- When recognised, R protein directly interacts with AVR protein, triggering defence response
- Defence response:
    - Synthesis of:
        - Antimicrobial enzymes, metabolites
        - Signalling molecules
            - Activate defence in near cells
    - Reinforcement of cell wall around infection site
- Detect and use fragments from cell walls of bacteria and fungi to recognise
- Plant genes → proteins that recognise pathogen proteins → making of enzymes, metabolites, activating defence in near cells, reinforcing cell wall around infection site

Basal Resistance:

- Also chemical barrier in passive defence
- Non-host resistance or innate immunity
- Pathogen recognition:
    - Plants have pattern recognition receptors (PRRs) on cell surfaces
    - PRRs can detect common microbial or pathogen-associated molecular patterns (MAMPs or PAMPs)
        - Are molecules shared by many types of microbes
            - E.g. Bacterial flagellin, Fungal chitin
    - Upon pathogen detection, gaps between cells close to reduce chance of pathogen spreading in tissue
- Plant cell surface receptors detect pathogen from its molecular patterns → pathogen detected → closing gaps between cells to stop pathogen from getting in

Hypersensitive Response (HR):

- Secondary line of resistance if basal fails
- Localised response
    - prevents the pathogen from
        - spreading to other parts of plant
- Plant cells in infected region produces reactive oxygen species (ROS) and other oxidative agents
    - Agents alter cell wall chemistry - trapping pathogen in host cell
- Cells around pathogen die, cutting off nutrient source of pathogen
    - Causes yellowing, dying leaves
- Local cell death (stop pathogen spreading)
- Changing of cell wall chemistry from ROS
    - to trap pathogen
    - ROS from infected plant cells

Systemic (throughout) Acquired Resistance:

- Broad spectrum, whole plant immune response
- Activated on initial, localised exposure to pathogen
- Combining of Salicylic Acid (SA) is essential for triggering SRA
- Produces Salicylic Acid (signal molecule) to alarm rest of plant tissues that pathogen has invaded
    - Plays role in plants memory of pathogen
- Communicates initiation of other chemical barriers
- Reduces severity of future infections from same pathogen

Fungal Pathogen:

- Disease: Myrtle Rust Disease ← Myrtle Rust Fungus ← Eucalyptus
- Symptoms:
    - Deformed leaves
    - Defoliation
    - Reduced Fertility
    - Stunted growth
    - Plant death in:
        - Bottlebrushes
        - Tea trees
        - Eucalyptus
- Method of Infection:
    - Spores land on stem - causing site of infection
    - Spores send out germ tube to find opening into plant

    - When opening found, fungus penetrates deep into plants tissue to access nutrients
    - Fungus uses nutrients to grow more fungi
- Plant Response:
    - Releases small protein molecules (effectors) into cell
    - Cell detects fungus receptors by releasing proteins acting as defence against fungus
        - Binds to effector proteins
    - Binds tells plant there is an infection
    - Plant knows about infected cells, sacrificing those around them
        - Kills them to cut off nutrient supply to fungus
    - After cell death → starvation - no nutrients → plant death
    - Gene-for-gene response: produces chitin - chemical that breaks fungus cell wall
    - hypersensitive

Viral Pathogen:

- Disease: Banana Bunch Disease ← Banana bunchy top virus ← banana
- Symptoms:
    - Prevents fruit production
    - Reduced growth:
        - Emerging leaves becoming choked and bunched
- Method of Infection:
    - Infected aphids and other insects tapping into phloem for food is vector
        - Creates pathway for transmission of virus
    - Leaf munching creates wound site for virus to enter into plant
    - Virus invades plant cells to replicate more viruses
    - Virus may inhibit cellular processes:
        - water and nutrient uptake
        - Photosynthesis
- Plant response:
    - Systemic acquired response
    - Hypersensitive response:
        - Yellowing leaves - evidence

Animal Responses to Pathogens

Lines of Defence:

First Line:

- Physical (skin)

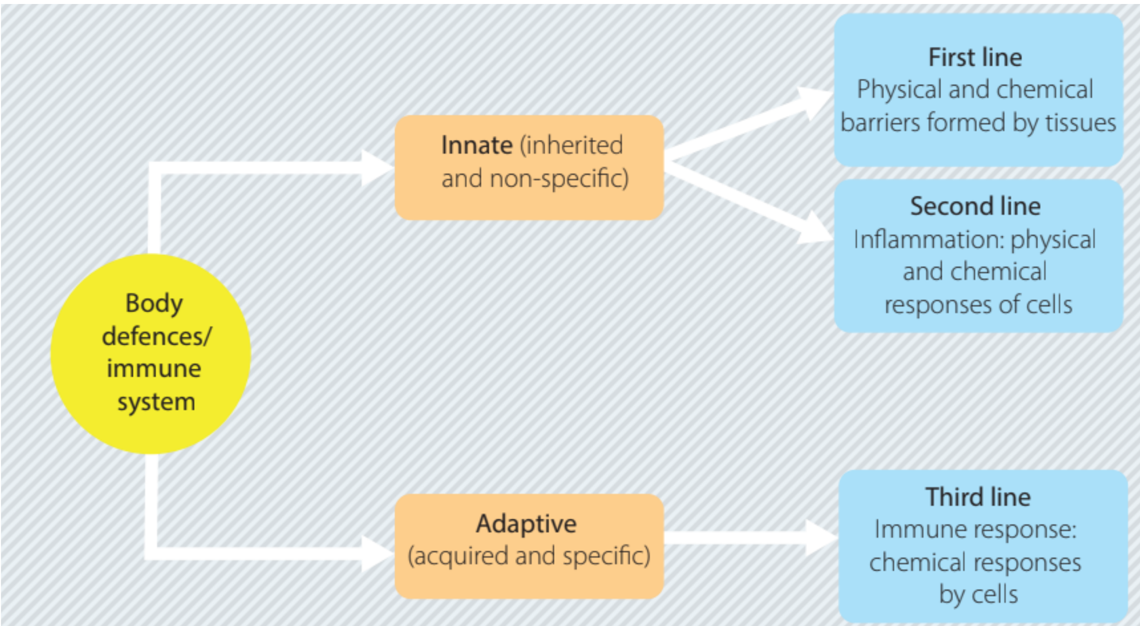
- Chemical (tears)
- Biological (sphincters) Barriers
- Non-specific:
    - Act on anything that is “not self”
        - (living organisms, organic, inorganic material)

Second Line:

- Non-specific chemical + cellular attack on pathogen
- Activated after pathogens pass first line
- Include:
    - inflammation
    - Phagocytosis
    - Fever (pyrexia)
    - Cell death to close pathogen formation (granuloma formation)

Third Line:

- Adaptive immunity
- Specialised cells that act if pathogen continues to invade



- Certain parts of the body must be sterile and free of contamination of microflora and pathogens
- Parts include:
    - Thoracic, cranial and abdominal cavities
    - Blood vessels of the cardiovascular system.

Immune System:

- Antigen:
  - any molecule that is foreign to body
  - Triggers immune response
- On surface of cells there are marker molecules
  - Identify cell as belonging to body ("self")
  - Protects body from attack from own immune system
- Pathogens have chemical markers: Antigens on surface
  - Immune system recognizes antigens as not belonging to body ("not self")
  - Antigens present activates immune response to destroy pathogens
- Any foreign cell, cell fragment, protein debris or toxin produced by pathogens have antigens

Lymphatic System:

- As blood moves around body,
  - Capillaries → Some plasma → tissues, becoming tissue fluid
  - Tissue fluid → lymphatic system
- Lymphatic system:
  - Has:
    - Lymph (a milky fluid)
    - Lymph nodes
    - Lymph vessels
    - Thymus
    - Spleen
    - Tonsils
    - Adenoids
  - Lymph vessels make one-way drainage system from all parts of body → point near heart
  - Cleaned lymph fluid → into blood through thoracic duct
  - Movement:
    - Muscles around vessels squeeze fluid in one direction
    - Valves stop fluid from moving back
- During infections:
  - Pathogens + their products may enter lymph fluid
  - Contaminated Fluid → local lymph nodes
    - Fluid filtered from substances:
      - Microbes
      - cellular debris
      - cancer cells

- Infections detected from enlarged local lymph nodes
- Site of infection may be detected by changes in lymph node:
    - Size
    - Shape
    - Texture

    As drainage routes are known for the node

White Blood Cells:

- Used in innate and adaptive immune responses
- Differ in:
    - Size
    - Having granules in cytoplasm (granular or agranular)
    - Color of granules and cytoplasm (pink - eosinophilic, blue - basophilic)
    - Shape of nucleus (lobed or mononuclear)
    - Secrete iron-containing enzymes to deal with pathogens
    -

Cell Type:	Characteristics:	Location:
Mast Cell	- Blood vessel dilation - Release of heparin and histamines - Recruitment of neutrophils and macrophages - Involved in allergic reactions	- Connective tissues - Mucous membranes

Macrophage	- Phagocytosis of:
	  - Pathogens
	  - cancer cells
	- Antigen-presenting cell
		- Blood vessels → tissues

Natural killer cell	- Kills:
	  - Tumour cells
	  - Virus-infected cells
		- Circulates in blood
		- Blood → tissues

Dendritic cell	- Antigen-presenting cell - Triggers adaptive	- Epithelial tissues in: - Skin

	immune response	- Lung - Tissues in digestive tract - Tissues → lymph nodes on activation

Monocyte	- Differentiates into phagocytic cells
	  - Dendritic cells
	  - macrophages
		- Stored in spleen
		- Spleen → infected tissues through blood vessels

Neutrophil	- Most common white blood cell at infection site
	- Releases toxins that:
	  - Kill bacteria and fungi
	  - Inhibit bacteria and fungi
	- Brings other immune cells to infection site
		- Blood vessels → tissues

Basophil	- Defence against parasites - Releases histamines that cause inflammation - Responsible for some allergic reactions	- Circulates in blood - Blood → tissues
Eosinophil	- Releases toxins that kill bacteria and parasites - Maintain homeostasis in body	- Circulates in blood - Blood → tissues

Pathogen ba

Microbiome:

- Microbiome: Microbes collectively in body
- Body gives:
  - Nutrients required for microbes
  - Living conditions for survival

- In return, microbiome:
    - Inhibits growth and multiplication of pathogens encountering body
- If conditions of body change:
    - Balance of microbiome changed
    - Growth and multiplication of pathogens uncontrolled
        - Leads to increase in their numbers
        - Leads to development of disease
    - An e.g. of a disease caused by imbalance is candidiasis (thrush)
- Natural balance may be upset by taking antibiotics to treat a bacterial infection
    - Antibiotics:
        - Reduce pathogenic bacteria in body
        - Reduce bacteria in natural microbiome
- Thrush:
    - Cause: Candida albicans (yeast)
    - Present in:
        - Mucous membranes:
            - in female genital tract
            - Mouth
            - Respiratory tract
            - Alimentary canal (digestive tract)
    - Numbers kept low by competition with other microbes in microbiome
    - Factors increasing numbers:
        - Suppression of immune system by:
            - Inherited conditions
            - Cancer treatment
            - Viruses (HIV/AIDS)
        - diabetes mellitus:
            - High sugar levels → candida overgrowth
        - Use of corticosteroids
            - Suppress immune surveillance
        - Hormones
            - High oestrogen levels - 2nd, 3rd trimesters of pregnancy
        - Oral contraceptives
            - Associated with higher oestrogen levels
        - General illness
            - Suppression of immune responses
        - Intravenous drug use

        - Like heroin

Physical Barriers against Infection:

Skin:

- Epithelial tissue
- 3 layers:
    - Outer epidermis
    - Underlying epidermis having:
        - Hair follicles
        - Sweat glands
        - Sebaceous glands
        - Nerves
        - Blood vessels
    - Lower hypodermis (subcutaneous tissue) (fatty layer)
        - Blood vessels
        - Lymph vessels
- Well supplied by blood
    - Provides wounds early access for:
        - white blood cells
        - Red blood cells
        - platelets
- Epidermis:
    - Keratin:
        - Waterproof protein
        - Extra layer of security against pathogens
        - Secreted by skin cells - Keratinocytes
        - Mechanically tough
        - Resistant to degradation by bacterial enzymes
    - Upper layer (stratum corneum):
        - Layer of flattened, dead skin cells in "brick and mortar" arrangement
            - Forms physical barrier against pathogens
            - As dead cells die and flake off (exfoliate), pathogens taken with them
    - When loss of integrity (completeness) of skin barrier, skin heals by:
        1. Inflammation:
        2. Proliferation: new cells rapidly multiplying to seal wound
        3. Maturation: cells mature, completing new barrier
    - Sebaceous and sweat glands in dermis:

- Produce:
    - Oil (sebum: range of lipid molecules)
    - Sweat
- Makes salty, acidic environment
- Prevents growth of pathogens

Mucous Membranes:

- Directly, continuously exposed to external environment
- Special type of epithelial tissue
- Number of layers and shapes of cells depends on location
- Typically pink
- Moist lining tissues
- Found in:
    - Digestive system
    - Respiratory system
    - Genitourinary system
- Features preventing pathogen entry:
    - Cell junctions between epithelial cells:
        - Anchors them together more effectively:
            - Increases cohesion
            - Prevents access to pathogens
    - Lined with hair-like cilia:
        - Beat in coordinated way to remove particles from respiratory system
            - Known as muco-ciliary escalator
    - Composed of sheets of cells:
        - Constantly growing, moving upwards to replace cells lost from wear and tear or pathogen attack
    - Secrete a lot of protective substances like:
        - Mucus
        - Lysozyme
        - Immunoglobulins (antibodies)

Tight Junctions:

- Blood vessels lined internally by endothelial cells
- Endothelial cells have special ways to adhere tightly to each other:
    - Helps prevent entry of pathogens from infected tissue → blood vessels
    - If pathogens → bloodstream, they can travel to distant sites and setup new points of infection

- E.g. Blood-brain barrier:
    - Formed by brain endothelial cells connected by tight junctions
    - Restricts diffusion of microscopic objects like bacteria into brain

Mucus:

- Slippery substance secreted by cells that line mucous membranes
- Coughing, sneezing out mucus expels microbes
- Body secretes mucus for:
    - Lubrication
    - Trapping debris, microbes
- Protects linings of body by trapping foreign substances:
    - Pathogens
    - Dust
    - Pollen
- Increased making of mucus in respiratory tract - sign of ill health
    - Made as body flushes away invading pathogens
- Digestive tract:
    - Prevents entry of pathogens through cells lining it
- Cervix:
    - mucus plugs guard entry to uterus by pathogens during pregnancy
- Intestines:
    - Goblet cells in small intestines epithelium secrete mucus
    - Mucus contains substances that inhibit replication of viruses (rotaviruses)

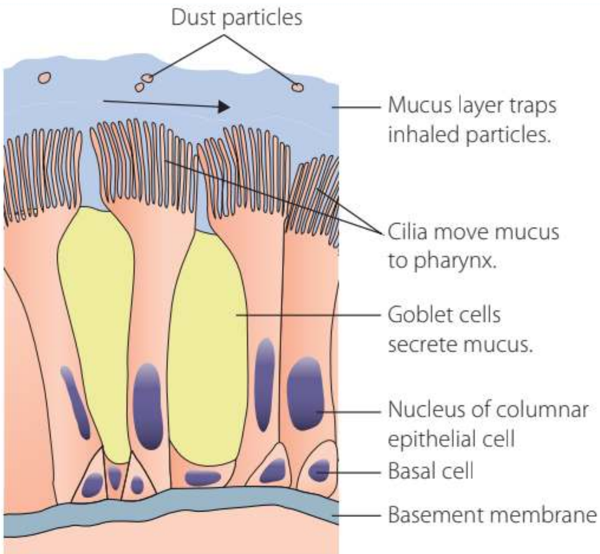


FIGURE 11.21 Mucus is produced by goblet cells that form part of the epithelial layer lining the respiratory tract.

Peristalsis:

- Occurs in the alimentary canal (digestive tract)
  - Mouth → anus
- Smooth muscle in digestive tract not under voluntary control
  - Acts automatically
- Peristalsis: Muscles contracts in coordinated way
  - Moves food in one direction only
  - Moves mucus and trapped microbes towards anus for excretion
  - Prevents faecal matter moving backwards into steriles parts of gut
- Lack of movement of small intestine - leads to intestinal bacterial overgrowth
  - Overgrowth occurs since bacteria have opportunity to reproduce

Sphincters:

- Circular muscles
- Retains constriction of:
  - Natural body passage
  - Orifice
- Relaxes as required by normal physiological functioning
- Found in:
  - Between oesophagus and stomach:
    - Lower oesophageal sphincter
  - Between stomach and duodenum:
    - Pyloric sphincter
  - Between ileum (small intestine - final part) and large intestine
    - Ileocaecal sphincter
  - Urethral sphincters:
    - Prevent release of urine from bladder unless done voluntarily
  - From common bile duct to duodenum:
    - Sphincter of the Oddi
      - One way flow of digestive juices from the common bile duct (gallbladder and pancreas)

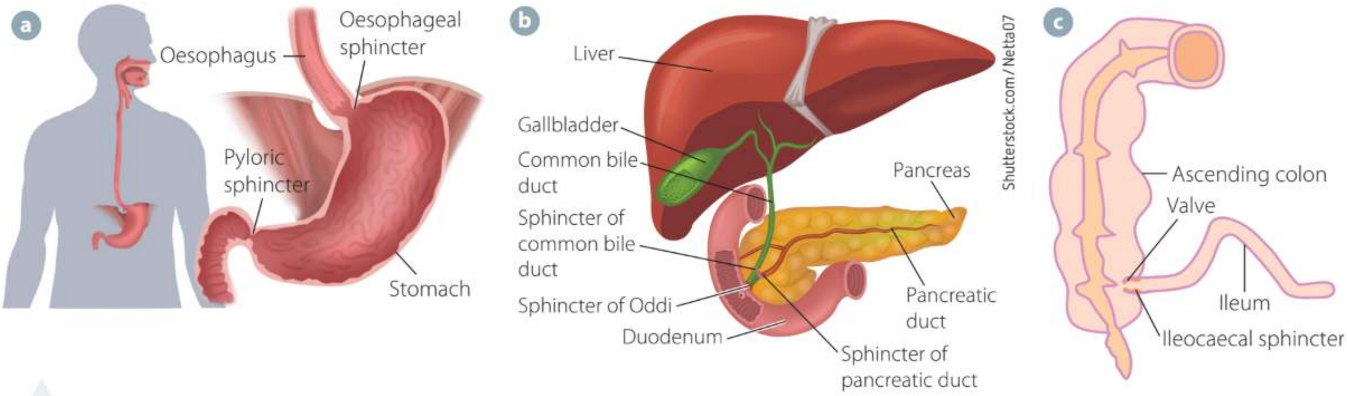


FIGURE 11.22 Sphincters help to physically seal off compartments in the body, to reduce the likelihood of pathogen invasion: **a** oesophageal and pyloric sphincters; **b** sphincter of Oddi and sphincter of pancreatic duct; **c** ileocaecal sphincter.

Physical Responses to Infection:

- Cells may die to seal off an area of tissue that is infected, not being defended by the body
  - Prevents infection from infecting other areas
- Granuloma: wall of dead cells forms capsule around infected tissue area
  - Cells inside it die, killing pathogens that infected them
  - Debris inside granuloma destroyed by macrophages surrounding “walled off area”
  - Formation is in 2nd line of defence
- Tuberculosis and leprosy bacteria form granulomas
  - Granuloma in lungs = “tuberoles”

Vomiting and Diarrhoea:

- Gets rid of pathogens and their toxins

Vomiting:

- Emesis
- Reflex action coordinated by vomiting centre (chemoreceptor trigger zone) of brain
- Response to signal of:
  - Presence of pathogens in gut (gastroenteritis)
- Expels harmful substances
- Hypersalivation happens before to:
  - Protect tooth enamel from stomach acid

Diarrhoea:

- Expels microbes quickly from gastrointestinal system

Increased Urination:

- Happens when bladder lining attacked by pathogen

- Responses:
    - Inflammation (cystitis)
    - Need to pass frequent small amounts of urine (pollakiuria)
- Help flushes out pathogens
- One-way action prevents pathogen moving into sterile areas (kidney, bladder)

Wound healing:

- Blood vessels may have been damaged = bleeding
- Priorities:
    - Stop bleeding to maintain normal blood pressure
    - Confront pathogens
        - Preventing infection
    - Heal, repair wound
        - Reestablishing barrier
- Stopping bleeding:
    - Blood vessels contract (vasoconstriction)
    - Platelet plug formed
    - Protein - fibrin forms mesh
        - Traps more platelets
        - Forms clot to seal wound
    - Inflammatory response starts

Chemical Defences against Infection:

- Non-specific chemicals secreted by epithelial tissue
    - Prevents pathogens from entering internal environment of body

Urine:

- Sterile until reaching lower urethra
- When animal or human is not urinating, pathogens may enter the lower urinary tract from:
    - Lower urethra
    - Penis
    - Vagina
- Females - shorter urethra than males
    - Greater risk of lower urinary tract infections
- Greater threats found in these areas:
    - Faecal bacteria (e. coli)
    - Commensal organisms (staphylococcus spp.)
- Bladder infection:

    - Microbes from urethra → bladder
- Flushing activity of urine keeps pathogens away from bladder
- Chemical components of urine defending against pathogens:
    - Antimicrobial peptides (AMPs):
        - Secreted by cells lining the urinary tract
        - Prevent binding of bacteria to epithelial cells
        - Break down (lyse) bacterial cells
    - pH of urine:
        - Range of 4.5 to 8
        - Average of 5 to 6 (slightly acidic)
        - Phagocytes work best when urine is alkaline
            - Doctors may prescribe urine alkalizers for urinary tract infections

Sebum and Sweat:

- Sebum: oily material
    - Secreted by sebaceous glands
    - Waterproofs skin
    - Lubricates skin
- pH of skin: 5.5 (acidic)
- Amino acids and fatty acids in sweat and sebum:
    - Because of lactic acid being present
- Lysozyme secreted in perspiration
    - Breaks down bacterial cell walls

Saliva:

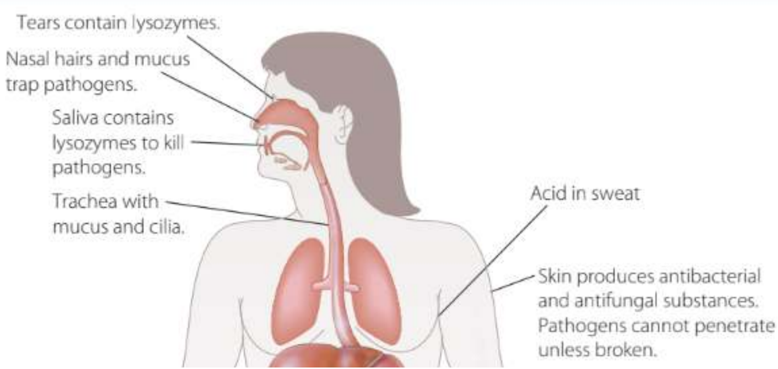
- Produced by salivary glands
- Complex mixture of:
    - Water
    - Mucus
    - Electrolytes
    - Enzymes, like amylase
    - Antimicrobial substances:
        - Lysozyme
        - Immunoglobulin A (Ig A)
- Has flushing action against microbes
- Chemical activity against microbes
    - Due to antimicrobial molecules contained in it

Tears:

- Produced by lacrimal glands
- Glands along eyelid edges
    - Secrete sebum-like substance that contributes to tears
    - Has antimicrobial properties
- Lacrimation: production of tears
    - Process produces tear film - covers cornea and conjunctiva
- Goblet cells:
    - In conjunctival epithelium
    - Secrete mucus
        - Help distribute film evenly
- Tear film chemical substances having antimicrobial properties:
    - Lysozyme, lactoferrin, lipocalin
    - Antimicrobial peptides (AMPs)
    - Complement
    - Immunoglobulin A (IgA)
    - Mucins

Gastric (Stomach) secretions:

- parietal cells lining stomach (gastric) wall secrete hydrochloric acid:
    - Creates highly acidic environment (pH 1-2)
        - Discourages growth, survival of microbes
- Pepsin:
    - Secrets by chief cells of stomach wall
    - Antimicrobial role thought to
- Bacterial pathogens can find a way around acidic conditions of stomach
    - By developing adaptive mechanisms allowing bacteria to survive
- Eating may raise stomach's pH above bacteria killing threshold
    - Bacteria may survive acidic stomach conditions, passing into intestinal tract
        - Causes gastroenteritis
- pH moves to 6 on release of basic bile into lumen of duodenum
    - Limiting factor of pathogen growth



Inflammation:

- Chemical response
- Helps wound repair
- Leads to pathogen destruction
- Five cardinal signs of inflammatory response:
  - Dolor: PAIN
    - due to increase of chemical mediators of inflammation
  - Calor: HEAT
    - due to increase in microcirculation
  - Rubor: REDNESS
    - associated with increased microcirculation
  - Tumor: SWELLING
    - as fluids move from intravascular space → extracellular space
  - Functio laesa: LOSS OF FUNCTION
    - due to pain and swelling
- Non-specific defence mechanism
- Occurs at site of infection
- When cells are challenged by pathogens or damaged in some way
  - Chemical "alarm signals" are released. Can be:
    - Histamines:
      - Trigger vasodilation and increase vascular permeability
        - Widening of blood vessel
        - Increase in blood vessel permeability
          - Allows white blood cells → site
    - Bradykinin
    - Serotonin
    - Prostaglandins (related to pain and fever of inflammation)
  - These chemicals:
    - cause capillaries to dilate:
      - Increasing blood flow to site of infection or injury
      - Cause area to become red, hot, swollen, painful, sometimes less mobile
    - Increase permeability of blood vessels
      - Allows certain white blood cells to move from blood → tissues to attack pathogens
        - Phagocytes:
          - Neutrophils
          - Macrophages
          - Dendritic cells

- Phagocytes are attracted to site of inflammation by chemotactic factors (chemical)
- Chemicals that increase body temp are released: endogenous pyrogens:
  - Inhibits growth of pathogens
  - Inactivates some enzymes, toxins made by pathogens
  - Increases rate of biochemical reactions happening in body
- When pathogens destroyed they are:
  - Removed along with toxins
- Tissues are then repaired
- When damage happens to tissues - wound is sterile:
  - Damage-associated molecular patterns (DAMPs) released by damaged cells:
    - Sends a chemical signal to tissue around to start inflammatory response
  - After, cellular replacement to restore damaged tissue

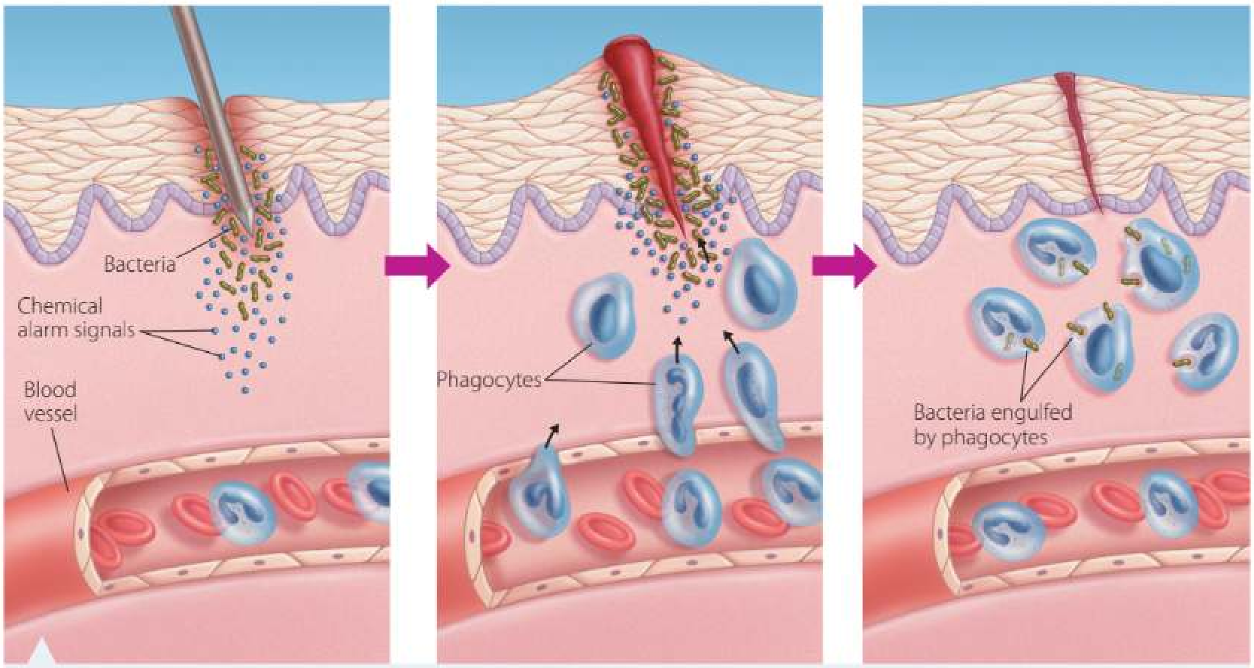


FIGURE 11.27 The inflammatory response: chemical signals at the site of bacterial entry attract white blood cells.

Phagocytosis:

- Response to chemical signalling
- Process by which phagocytes change their shape
  - To surround foreign particle (bacterium) and fully enclose it
- When particle is inside cell
  - Enzymes destroy it
- Phagocytes: specialised blood cells to leucocytes
- Main types phagocytes:
  - Neutrophils
  - Monocytes MONOCYTEs
  - macrophages

    - Dendritic cells DENDRITIC CELLS
    - Natural killer cells
- Not always successful
    - Pathogens may repel phagocytes
    - May escape before being destroyed
- In severe infections:
    - Immature neutrophils show toxic changes ("toxic neutrophils") when seen in blood smear
        - Due to bacterial toxins circulating in blood (sign of septicaemia)

Neutrophils:

- All phagocytic cells + neutrophils come from bone marrow
- Can deform and squeeze between endothelial cells lining local capillaries
    - To move from blood → tissues (diapedesis)
- Are the 1st to move to site of infection to inactivate pathogens
- Short acting
- Self-destruct after a few days
- Used by body to fight short, severe infections
- Increase in neutrophils circulating around body
    - Indicative of inflammation somewhere
- Complex series of electron and ion flows occur:
    - Movement of ions into vacuole of neutrophil
        - Creates conditions for death of microbes
        - Dead microbes digested by enzymes like proteases

Monocytes:

- Circulate in blood
- then attract to inflamed tissue
- Move from Capillary walls → inflamed tissue
    - Transform into MACROPHAGES + DENDRITIC CELLS
        - Both antigen presenting cells
        -
- Long-lasting phagocytes - stay in tissues or travel from blood vessels → infected tissues
- Used by body to fight chronic (long-lasting) infections
- After killing of foreign particle
    - Parts of antigen are displayed on surface of macrophage or dendritic cell (antigen presentation)
- When tissues damaged - monocytes recruited to tissue from blood
- On surface of monocytes - toll-like receptors (TLRs):

- Recognise pathogen associated molecular patterns (PAMPs) from bacterial cells
- Quickly differentiate to macrophages and dendritic cells
    - Remove microbes, lipids, dying cells using phagocytosis
- Dendritic cells - antigen presenting cells
    - With macrophages, act as bridges between innate and adaptive immune systems
- May also differentiate into osteoclasts ("bone breakers"): cells responsible for maintenance, repair and remodelling of bone

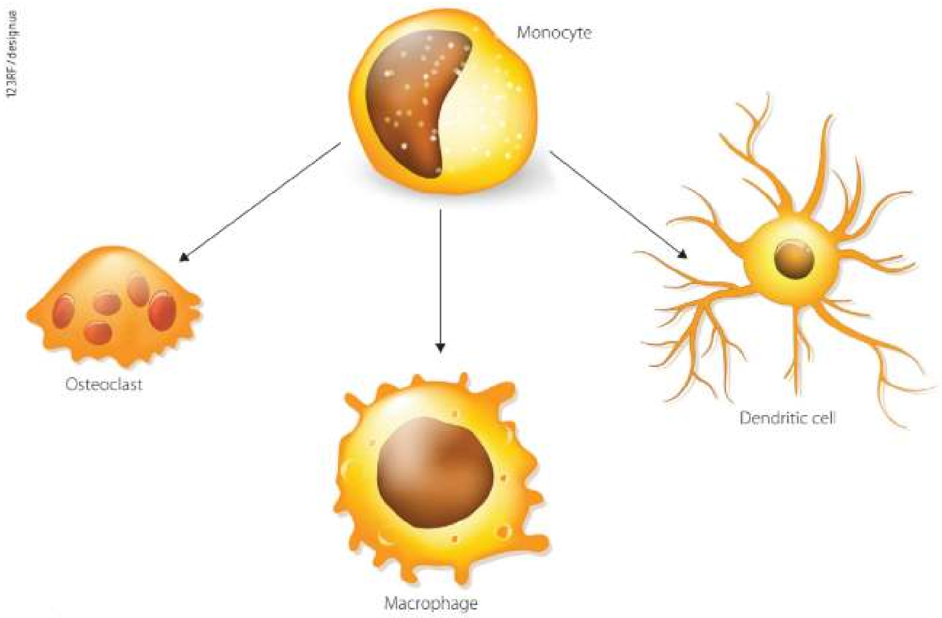


FIGURE 11.31 Monocytes are capable of transforming into three different cell types.

The Complement System:

- A group of around 20 soluble proteins
- Assist other defence mechanisms in killing extracellular pathogens
- Stimulates phagocytes to become more active
- Attracts phagocytes to infection site
- Destroys membranes of pathogen
- Made in:
    - Liver cells
    - Macrophages
- Circulate in blood
- Part of initial response to infection
- Pathogens (with antigens) bind to antibodies:
    - Complement proteins attracts to and coats the antigen-antibody complex
    - Signals for phagocytes and lymphocytes (B) to kill pathogen
- Some proteins punch holes in microbial cell membranes

- Cell contents leak out
- Part of innate immune response

Fever:

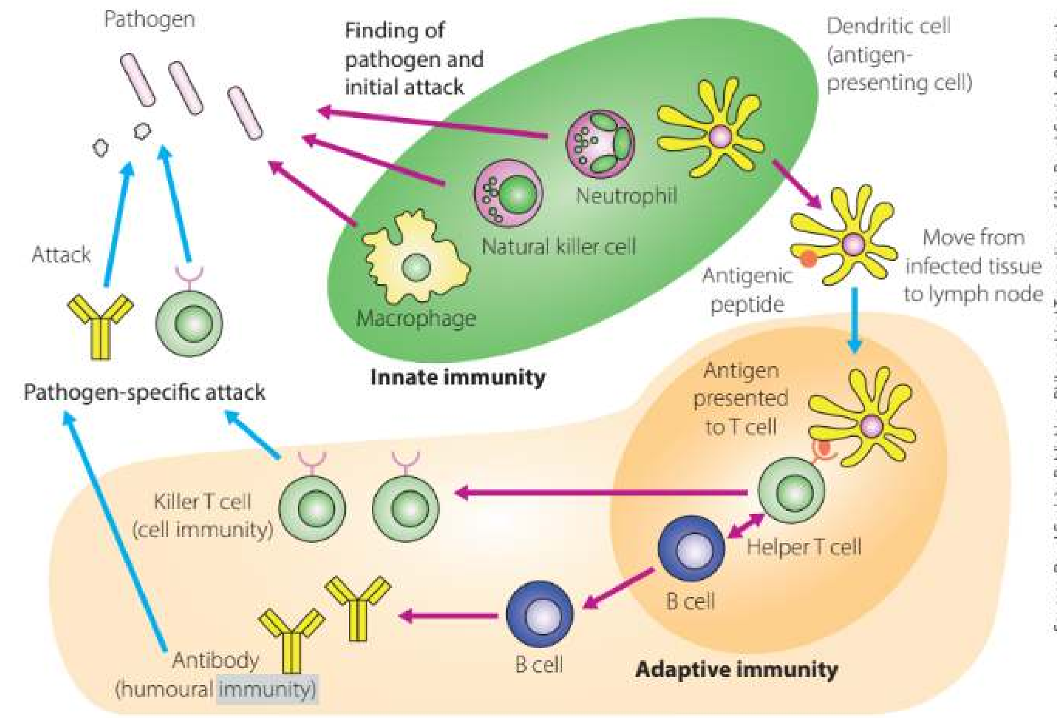
- Hypothalamus - has cells that regulate body temp
- Body may react to pathogens by letting tissue heat up
    - Done by pyrogens (“fever-causing chemicals”, “fire-starters”)
- Purpose is to kill, prevent growth of pathogens
    - Heat denatures the enzymes in pathogens
        - Limiting their functions (reproductive, food taking)
        - Eventually killing them
- May enhance activity of white blood cells
- “Pyrexia”
- Comes with: sweating, chills, muscle aches, general weakness
- Temporary response to pathogen is 2-3 day fever, anything very high for a long time should be medically assessed
    - Unexplained fever in child younger than 3 months - doctor

Cytokines:

- Chemical messengers
- Produced during infection
- Promote development and differentiation of T and B lymphocytes for 3rd line of defence
- E.g. Interleukin (IL)
- Form a link between innate and adaptive immune systems
- Burst of cytokines occur when infected cells signal nearby uninfected cells to release cytokines
- E.g. interferons
    - Act as chemical signal to viruses to stop replicating
        - Done by:
            - signalling uninfected cells to destroy RNA
            - Reduce protein synthesis
    - Infected cells signalled to do apoptosis (programmed cell death)
- Role in feeling lethargy, muscle pain, nausea when experiencing infection
    - Implication - animal isolates itself and rests, preventing spread of infection to other animals

Human Immune System:

- Innate and adaptive immune systems:
    - Only found in vertebrates with jaws



- Innate immune response:
  - Genetically preprogrammed
  - repeated exposure to same pathogen
    - Body reacts same way
  - Present from birth
  - Does not alter
    - Only changes in poor health due to other problems
  - Involves:
    - No learning
    - Memory formation

      By the body
- Adaptive immune response:
  - 3rd line of defence or acquired immunity

Adaptive Immune Response:

- 2 arms:
  - Humoral response:
    - Effective against pathogens in body fluids
  - Cell-mediated response:
    - Effective against intracellular pathogens (inside cells)
- B and T lymphocytes:
  - Generate adaptive immune response
  - (B and T cells)

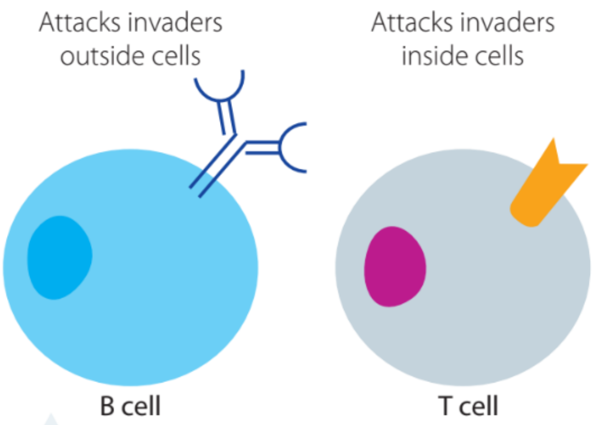


FIGURE 12.3 B and T lymphocytes target different types of antigens.

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B Lymphocytes:

- Made in bone marrow
- Mature in bone marrow (B)
- Some develop into plasma cells
  - Produce antibodies (immunoglobulins) against pathogen
- Some develop into memory B lymphocytes:
  - Give immunological memory
  - Long term defence against antigens
  - If organism exposed to same pathogen again:
    - Recognised by memory B cells
    - Memory cells divide into plasma cells (producing antibodies)
    - Immune response is:
      - Faster
      - Stronger
      - longer-lasting

Antibodies:

- Protein molecules with specific molecular structure
  - Helps them to recognise and bind to specific pathogens
  - Aka Humoral response
    - Humoral response occurs in Bodys:
      - Blood
      - Tissue fluids
- Don't destroy antigens or pathogens
  - Instead interfere with functioning of pathogen by making:

        - pathogen unable to cause damage
        - Easier for other parts of immune system to destroy it

Strategies to Destroy Pathogen:

- Neutralisation:
    - Deactivating a pathogen or toxin
        - By blocking its active site
- Precipitation:
    - Antibodies bind to soluble antigens:
        - Cause them to form insoluble clumps
        - Can then be eaten by phagocytes (phagocytosis)
- Agglutination:
    - Antibodies bind to antigens on surface of cells
        - Form clumps of cells
        - Phagocytes → eat them
- Activating Complement System:
    - Helps:
        - Disarm pathogens
        - Increase phagocytosis
        - Increase inflammation
        - Cell lysis → pathogen removal

T Lymphocytes:

- Made in bone marrow
- Mature in thymus (T)
- Aka cell-mediated response
- Different from first 2 lines of defence:
    - Is specific
    - Involves great diversity of possible responses
    - Has memory
    - Capable of self-tolerance

The Specific Response:

- Antigens:
    - Consist of:
        - Bacteria
        - Viruses
        - Fungi

        - Worms
        - Tumour cells
        - Other non-cellular materials
    - Generally molecules made of:
        - protein or polysaccharides found on:
            - cell membranes
            - Viral coats
    - Antigen molecules have regions known as epitopes that can:
        - Be recognised by specific lymphocytes
- Specific lymphocytes:
    - Recognise specific chemistry of each epitope
    - Then can target these antigens
- B and T lymphocytes:
    - Have complementary antigen binding sites on cell membranes
        - Aka antigen receptors
    - Epitopes may be bound by:
        - cell receptors on B and T lymphocytes
        - Or by free antibodies in extracellular fluid
- Antibodies:
    - Are Y-shaped molecules that:
        - Consist of 4 chains of protein:
            - 2 large heavy chains
            - 2 small light chains
    - 5 classes of antibodies - 5 types of heavy chains:
        - IgA
        - IgE
        - IgG
        - IgM
        - IgD
    - Specific region of molecule:
        - Binding site for antigens using epitopes
    - Has 2 binding sites
        - Each binding site specific for particular antigen
    - Pathogens may have a few different epitopes on surface
        - They bind to different antibodies
    - Antigen-antibody complex:
        - Place where antibodies and antigens bind

- Antigen Receptors:
    - B cell:
        - On surface of cell membrane
        - Have same affinity for a specific pathogen as antibodies that this cell will secrete when turned into plasma cell
    - T cell:
        - Surface proteins
        - Structure not like antibodies
        - Not secreted by cells
        - Are bound into cell membrane

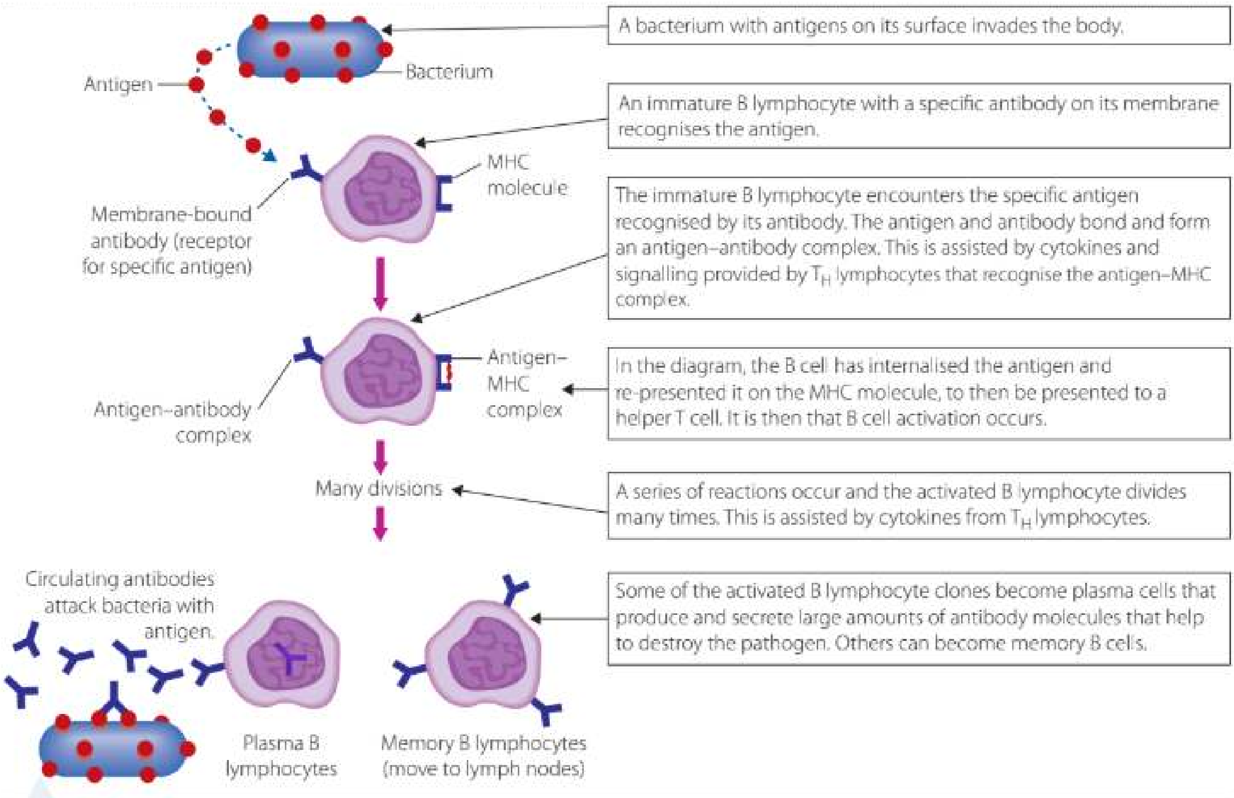
Pathogen Diversity:

- T and B cells have 1000s of antigen receptors on cell membranes
- For 1 T or B cell:
    - Each of the receptors is capable of recognizing one of all possible antigens in world
- Many combinations of receptor types to human immune system
    - Combinations determined from persons genes
    - set from birth
- Clonal Selection Theory:
    - States that all B cells for all possible antigens are present
        - In small amounts in immune system at birth
    - When pathogen antigen present in body:
        - B cell specific for antigen is:
            - Activated
            - Then cloned
        - When antigen destroyed:
            - Cloned B cells remain
                - Ready for next time when antigen is in body
                - Become memory B cells

Primary Immune Response:

- When adaptive immune system first exposed to pathogen
    - Response by B and T lymphocytes (cells):
        - known as primary immune response
    - Cells produced to clear infection:
        - plasma cells
            - Produce antibodies
        - Cytotoxic T cells

- Body stores these cells for next time it finds same pathogen in body
- Person then possibly immune to infection by pathogen

- 

- Response is fast, short-lived (not long lasting)
- Total number of antibodies produced in first response < in future infections
- Adaptive immune system in future infections when body exposed to pathogen:
  - More rapid response
  - More antibodies made
- Aka secondary immune response
- Is reason why booster vaccines are recommended for many diseases
- Secondary immune response happens because:
  - Of memory T and B cells produced from 1st infection from pathogen

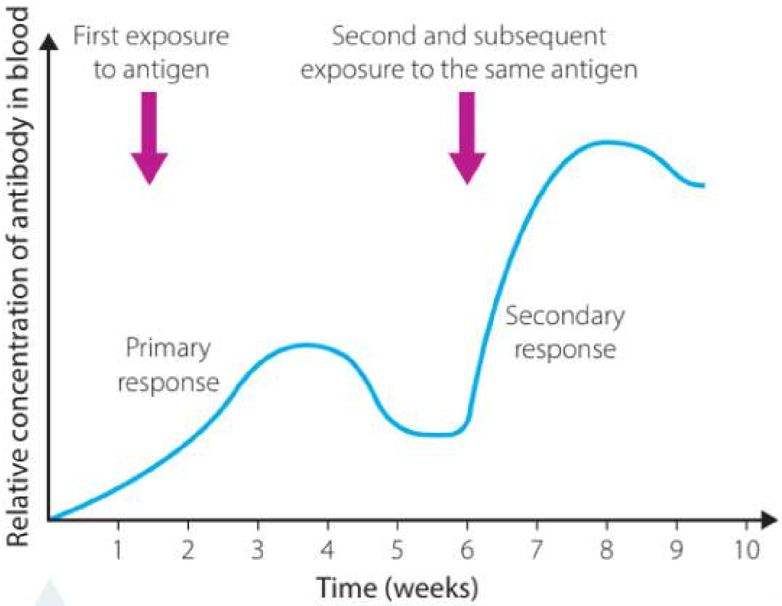
- 

FIGURE 12.8 The primary and secondary immune responses

How B and T cells don't attack body:

- B and T cells learn to not harm body in which they develop
- When cells mature in bone marrow, thymus:
    - Are examined for tendencies to harm body
- Cells that have genetic tendency to harm body
    - Are:
        - identified
        - Then neutralised
- These cells never mature and harm body
- The only cells that stay there are:
    - those that react to
        - "Other"
        - "Non-self" antigens
- Autoimmune disease may happen if this process not fully effective

Humoral Response to Pathogens in Body Fluids:

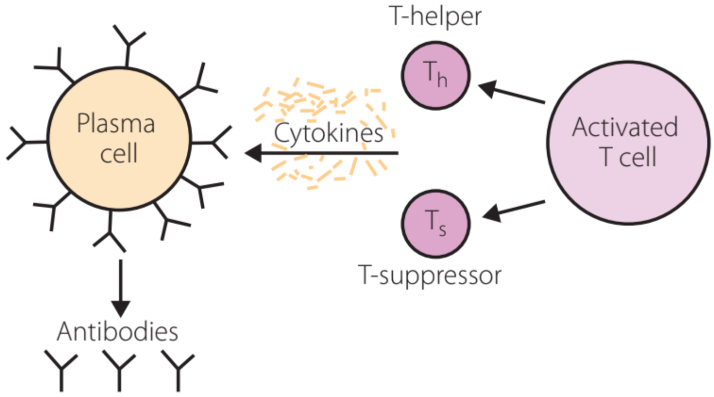
- Humoural response:
    - Used for first exposure to pathogen outside body cells
    - B lymphocytes mainly responsible
- Mature B cells
    - stored in:
        - lymph nodes
        - Peripheral lymphoid tissues

- Circulate in blood
- Lymph nodes:
  - Glands located at
    - points around body where
      - Tissue fluids drain from major sites
- Helper T cells:
  - Release cytokines - involved in activation of B cells
    1. B cell multiplies:
       - Making many copies of itself
         - With same specificity as original B cell
    2. B cells differentiate into 2 cell types:
       - Plasma cells:
         - Release antibodies (antibody factories or effector cells)
         - short-lived
       - Memory cells:
         - Stored for future infections
         - Long-lived

B Lymphocytes (cells)

- Protein-containing antigens:
  - T cells:
    - Are:
      - activated
        - Bind to pathogen
    - Release cytokines (special molecules)
      - E.g. interleukin-2 (IL-2)
  - When b cells exposed to IL-2 they:
    - Multiply rapidly
    - Differentiate into:
      - Plasma cells
      - Memory cells

FIGURE 12.10
Activated T cells assist in the activation and transformation of B cells into plasma cells.



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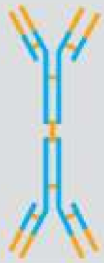
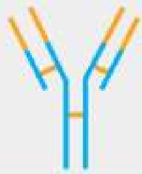
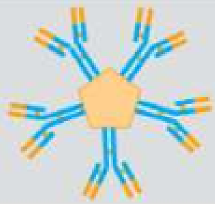
- Long polysaccharide chain antigens:
  - Bind directly to B cells
    - Cause them to
      - Multiply
      - Differentiate into plasma cells
  - No memory cells made
  - These antigens known as thymus-independent antigens

Antibodies:

- Molecules
- Produced by plasma cells
  - In response to exposure from specific pathogen
- Not all antibodies the same
- Not found in equal concentrations in body
- Aka immunoglobulins (Ig)
- 5 classes:

NAME	PROPERTIES	STRUCTURE
IgA	Found in mucus, saliva, tears and breast milk. Protects against pathogens.	
IgD	Part of the B cell receptor. Activates basophils and mast cells.	
IgE	Protects against parasitic worms. Responsible for allergic reactions.	
IgG	Secreted by plasma cells in the blood. Able to cross the placenta into the foetus.	
IgM	May be attached to the surface of a B cell or secreted into the blood. Responsible for early stages of immunity.	

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Cell Mediated Response to Intracellular Pathogens:

- Antibodies have no access to pathogens inside cells
- Cell-mediated immunity:
  - T lymphocytes:
    - Target
    - Destroy

      Whole infected host cell with pathogens in them
    - Make direct contact with infected cells using special receptors
      - Can:
        - Identify
        - Destroy

          Tumour cells
    - Control cell-mediated immunity
- Effective against:
  - Protozoa

- bacteria

  inside cells
- Some fungi
- Cancer cells
- Transplanted tissues

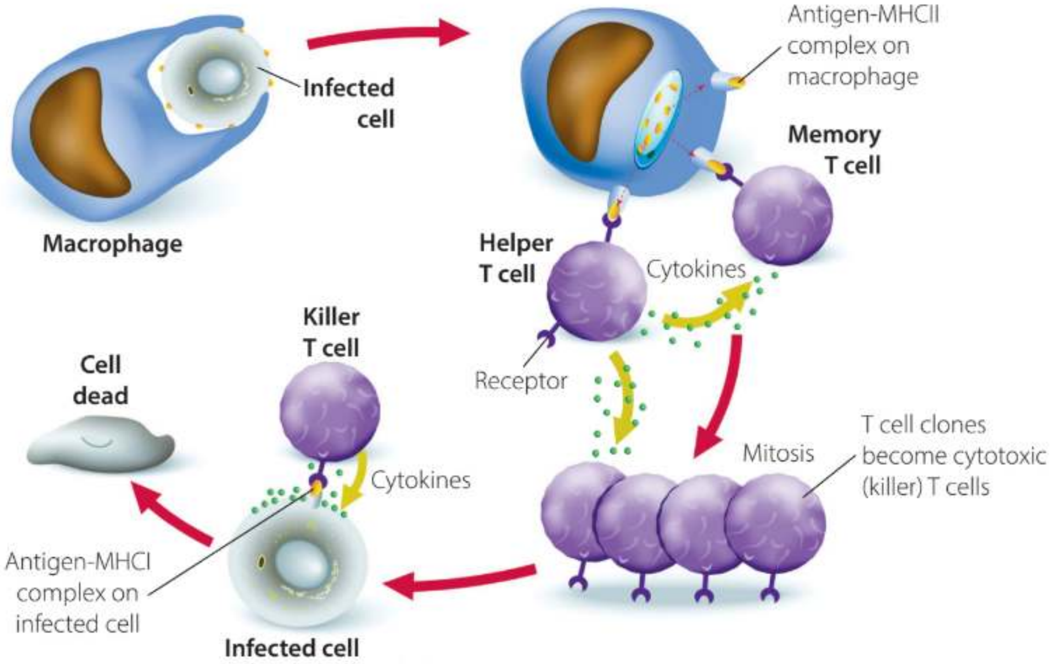
- 

FIGURE 12.12 The cell-mediated immune response

T Cells and MHC Molecules:

- T cells recognise fragments of antigens
  - May be antigens that:
    - have been partly digested by macrophages
    - Partly digested by invaded cells
- Kind of digested antigens:
  - Carried to
  - Displayed on cells surface
    - By special protein molecules
      - Major histocompatibility molecules (MHC)
- MHC molecules:
  - “Self-identity” molecules of organisms
  - +100 genes that code for
    - an individual's MHC marker molecules
  - Each person has
    - different set of MHC marker molecules on their cells
  - Have binding site for specific antigen

- 2 MHC classes:
  - MHC 1:
    - In all nucleated cells in vertebrates
      - Including platelets, excluding red blood cells
    - Infected cells represent a fragment of antigen
      - Bound to an MHC 1 molecules
  - MHC 2:
    - In:
      - Antigen-presenting cells:
        - Macrophages
        - Dendritic cells
        - Activated B cells
    - Macrophages present fragment of antigen bound to MHC 2 molecule
      - Known as antigen-MHC 2 complex
- T cells:
  - Produced in primary lymphoid tissue
    - Of bone marrow
  - Mature in thymus gland
    - In thoracic (chest) cavity
  - After maturing:
    - T cells released into blood
    - Migrate to:
      - Spleen
      - tonsils
      - Peyers patches of intestines
      - Lymph nodes

Types of T Cells:

Helper T Cells:

- Secrete cytokines
  - E.g. interleukin 2
    - Activate both adaptive immune responses
      - Humoural
      - Cell-mediated
  - Some cytokines released:
    - Stimulate activity of macrophages
- Promote other immune responses:

    - Increase activity of phagocytes
    - Promote inflammation
    - Stimulate production of killer T cells (IL-2)
    - Stimulate B cells (IL-2) to differentiate into:
        - Plasma cells
        - Memory B cells
    - Activation of B and T cells using IL-2
        - act as bridge between B and T cell parts of adaptive immune system
- Have a protein receptor that recognises
    - MHC 2 molecules bound to antigen fragment on surface of phagocytes (antigen presenting cells)

Cytotoxic T Cells (Killer T Cells):

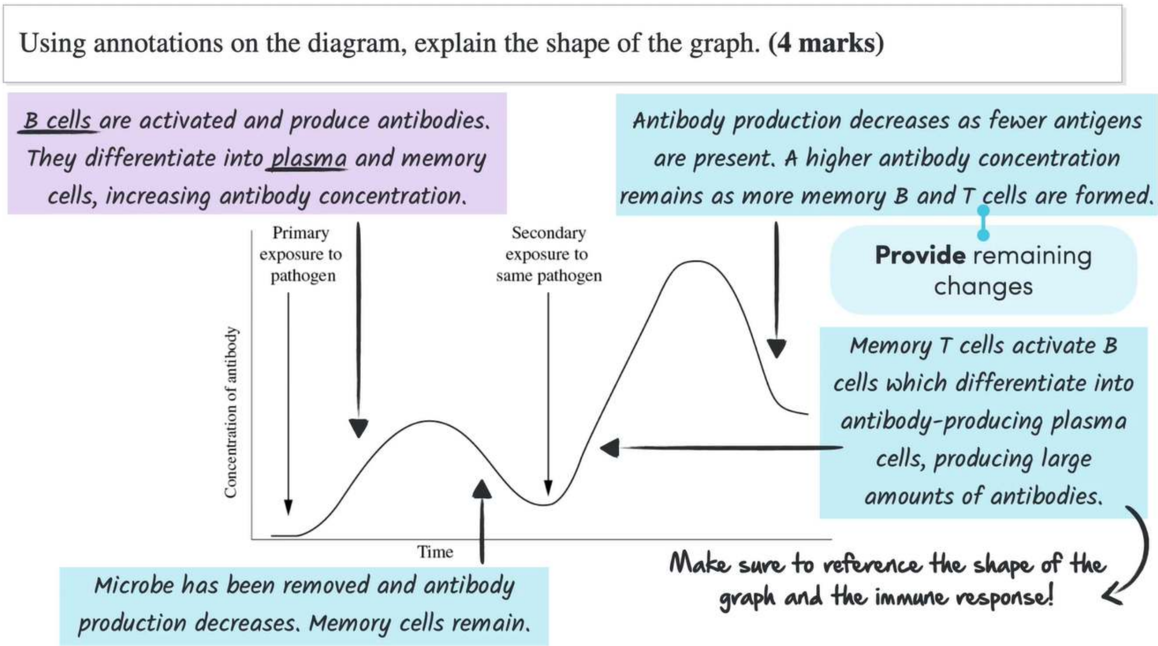
- Activated by helper T cells
    - They respond by producing many copies of themselves
        - Form an "army" of identical killer cells
            - Move to site of infection
            - Bind with infected cells
            - Release chemicals (cytokines)
                - Destroy infected cell
- Find infected body cells
- Binds to them and destroys them
- Kills:
    - Foreign cells
    - Infected cells
    - Abnormal cells
- Use surface receptors to recognise MHC 1 antigen-antibody complex
    - Found on cells infected by pathogen
- Production of cytokines
    - Cause lysis of infected cell

Suppressor T Cells:

- Turn off immune system after
    - Antigen is
        - Contained
        - Destroyed
        - Or removed

Memory T Cells:

- Make clones of themselves
  - Remain in body to
    - Respond to infection in future
- Give body long-term defence against pathogen
- Stay there after infection
  - Enable a larger, faster response when reinfected w same pathogen
- When body → same antigen:
  - Memory T cell recognise it
    - Divide into
      - Killer T cells
      - Helper T cells



- Include **WHY** antibody concentrations increases or decreases

Summary of Cell Mediated Immunity:

- Carried out by cell-mediated immunity

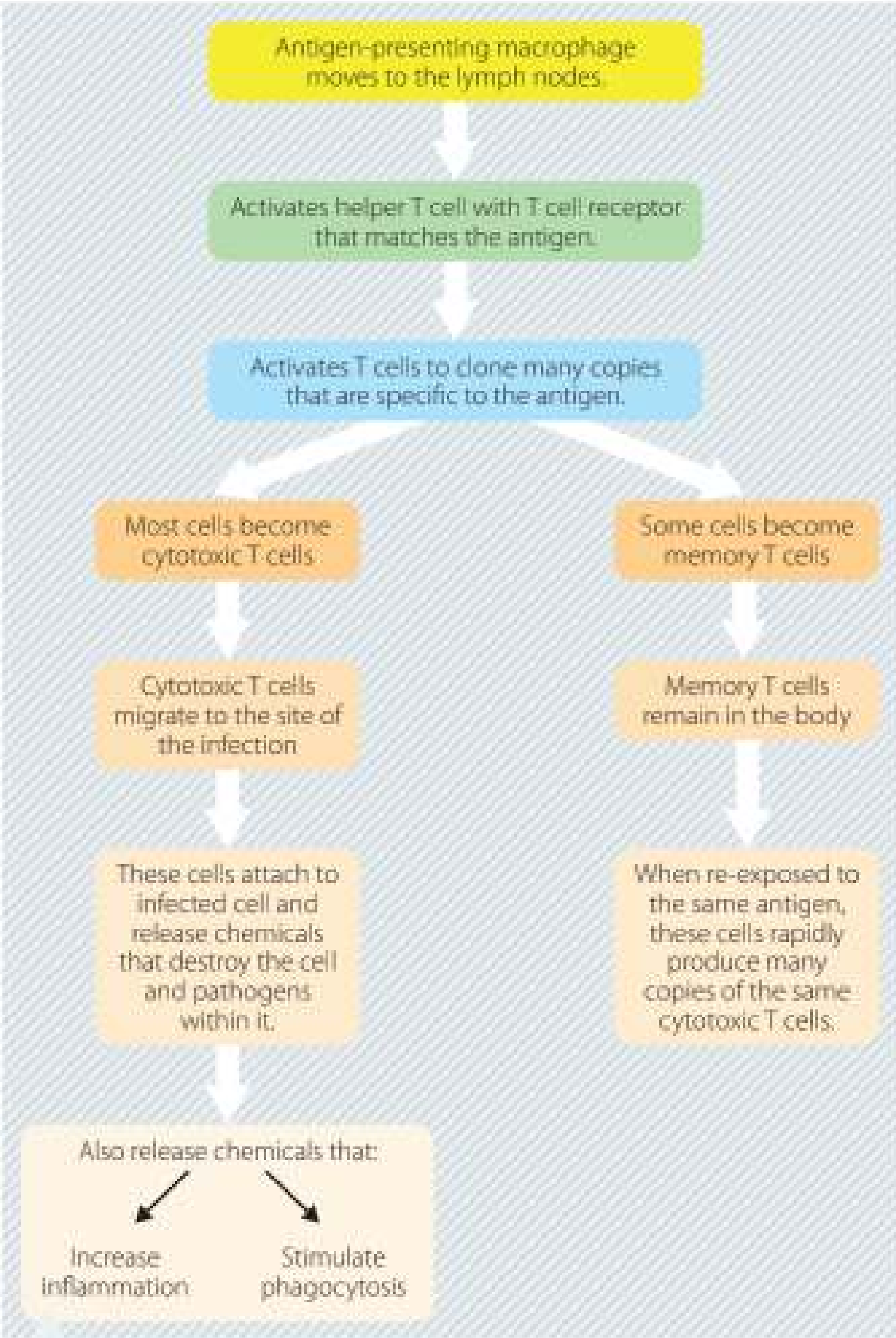


FIGURE 12.14 Steps in the process of cell-mediated immunity

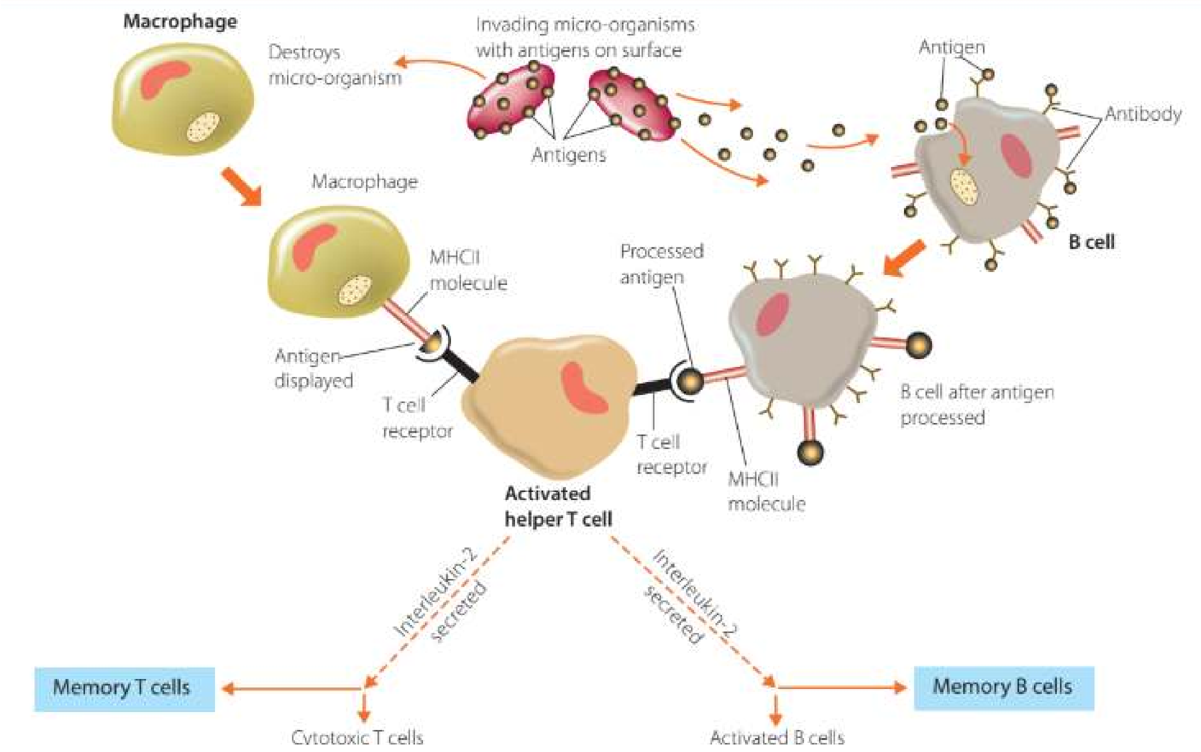
1. Foreign material engulfed by macrophages
   - Then displayed on them by attaching them to MHC 2 molecules
2. Macrophages move to lymph nodes:
   - Inspected by helper T cells that have T cell receptor
     - T cell receptor corresponds to antigen being presented
3. Helper T cells activate
   - Specific for antigen, cloning of millions of
     - cytotoxic T cells
     - memory T Cells

4. Killer T cells leave lymph nodes
    - Migrate to infection site
        - Antigen receptors bind with antigen displayed on infected cell
5. Killer T cells release cytokines
    - Destroy cell and pathogens in it
6. Cytokines increase inflammation, attract more macrophages
    - Carry out phagocytosis
        - Help destroy pathogens
        - Clear any debris
7. Some cytotoxic T cells produce chemical (interferon)
    - Protects healthy cells around infected cell from viral invasion
8. Once infection defeated
    - Suppressor T cell release other chemicals
        - To stop making and action of killer T cells
9. Memory T cells made at the time and specific to that antigen
    - Stay in body in lymph nodes
    - When re exposed to same antigen containing pathogen,
        - rapid production of same killer T cells
            - Prevents body from developing symptoms of disease again

Interaction of B and T cells in the Adaptive Immune System:

- macrophage encounters foreign particle with antigen on its surface
    - Surrounds and engulfs it
    - Phagocytosis
- During process of destroying particle
    - Antigen that was present on particle surface
    - Moved to surface of macrophage
    - Transports antigen to lymph nodes
- Antigen presenting macrophage
    - Presented to helper T cell
        - Has T cell receptor matching to specific antigen
    - Helper T cell then activated
- Helper T cell can be activated by
    - B cell
- When B cell finds antigen that matches to its surface antibodies
    - Binds antigens to its antibodies
    - Processes the antigen

- Attaches it to B cell mhc 2 molecules
- Presents antigen on mhc 2 to helper T cells with
    - Matching T cell receptors
- Cytokines - chemical signals
    - Secreted by helper T cell to
        - Activate more of the same
            - Helper T cells + macrophages
- Interleukin-2 (IL-2) (cytokine) produced by helper T cell
    - Activates production of:
        - Clones of B cells specific to antigen
        - Clones of killer T cells
            - Having that particular antigen receptor on their surface
- When immune response has defeated the infection
    - Suppressor T cells responsible for:
        - Suppressing the activity of:
            - B cells
            - Killer T cells



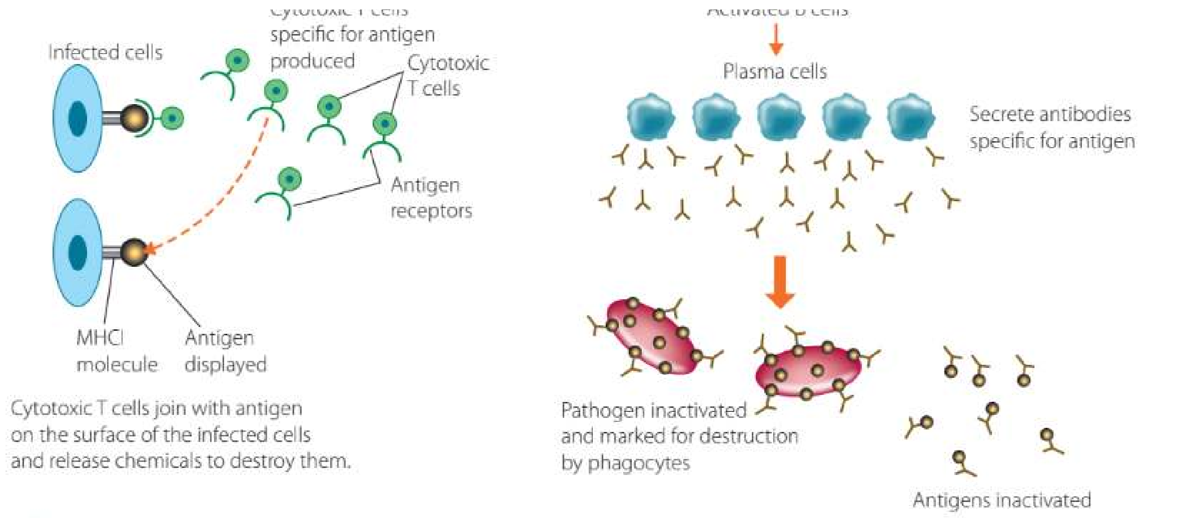


FIGURE 12.16 The immune response, showing interactions between B cells and T cells

The Immune Response after Primary Exposure to Bacteria:

Innate Immune Responses:

- Complement proteins puncture the bacterial
  - Cell wall
  - Membrane

  Make it susceptible to osmotic lysis
- Opsonized bacteria
  - (covered with antibodies and complement proteins
  - Are phagocytosed
- Neutrophils are active
  - Peripheral blood levels of neutrophils increase
    - In response to:
      - Increased bone marrow production
  - Extreme infections:
    - May cause neutropenia
      - Due to increased demand for neutrophils
        - Demand exceeds ability of bone marrow to supply them
  - Monocytosis (increased monocyte count)
    - Happens in mostly solving of bacterial infection

Adaptive Immune Response:

- Response depends on if bacteria is
  - Intracellular
  - Extracellular

Intracellular Bacteria:

- Cell-mediated immunity launched against
    - Intracellular bacteria
        - E.g. salmonella
        - Can't be found by
            - Complement
            - Antibodies
- Infected macrophages present
    - Bacterial proteins on cell surface
        - Using MHC 2 receptors
- Helper T cells detect these
    - Release interferon
        - Stimulates macrophage
            - To digest bacteria-infected macrophage
- ↑ e.g. of innate and adaptive systems working together
    - at same time to solve same problem

Extracellular Bacteria:

- E.g. staphylococcus aureus
- Most frequent of all
- Protection mechanisms related to
    - hosts natural barriers
    - Other innate immune responses
    - Antibody production
        - By adaptive immune system
-

Natural barriers	- structures/substances such as
	  - Skin
	  - Mucous membranes
	  - Sebum
	  - Urine
	  - Gastric acid

Innate immunity	- Complement proteins
	- Phagocytes
	  - Natural killer cells
	  - Neurophils

	- Macrophages - Dendritic cells
Adaptive immunity	- Antibody making by plasma cells - Cytokine making by T cells - + killer T cell activation - Memory B cells made

Immune Responses to Vibrio Cholera - Case Study:

- Causes profuse diarrhoea and vomiting
- Death caused by
  - Resulting severe dehydration and shock due to fluid loss
- Bacteria that cause cholera
  - Is a curved or comma-shaped (vibrio)
  - Gram-negative rod
- Human immune system responding to primary exposure:
  - Innate response:
    - making of cytokines like interleukin
    - Neutrophils → gut lining
    - Making of antibacterial peptides
      - E.g. defensins
      - Kill bacteria directly
      - Maybe alter host gene expression to induce more cytokines
        - To induce inflammation
  - Adaptive response:
    - IgA (antibody class) secreted by gut lining
      - To protect from bacterial colonisation
    - After a week there is an increase in
      - Circulation of B lymphocytes specific to v cholera antigens
    - Serum antibody peaks at 1-3 after first infection
      - Decreases after about a year
    - B memory cells form
      - Provide secondary response with same bacteria

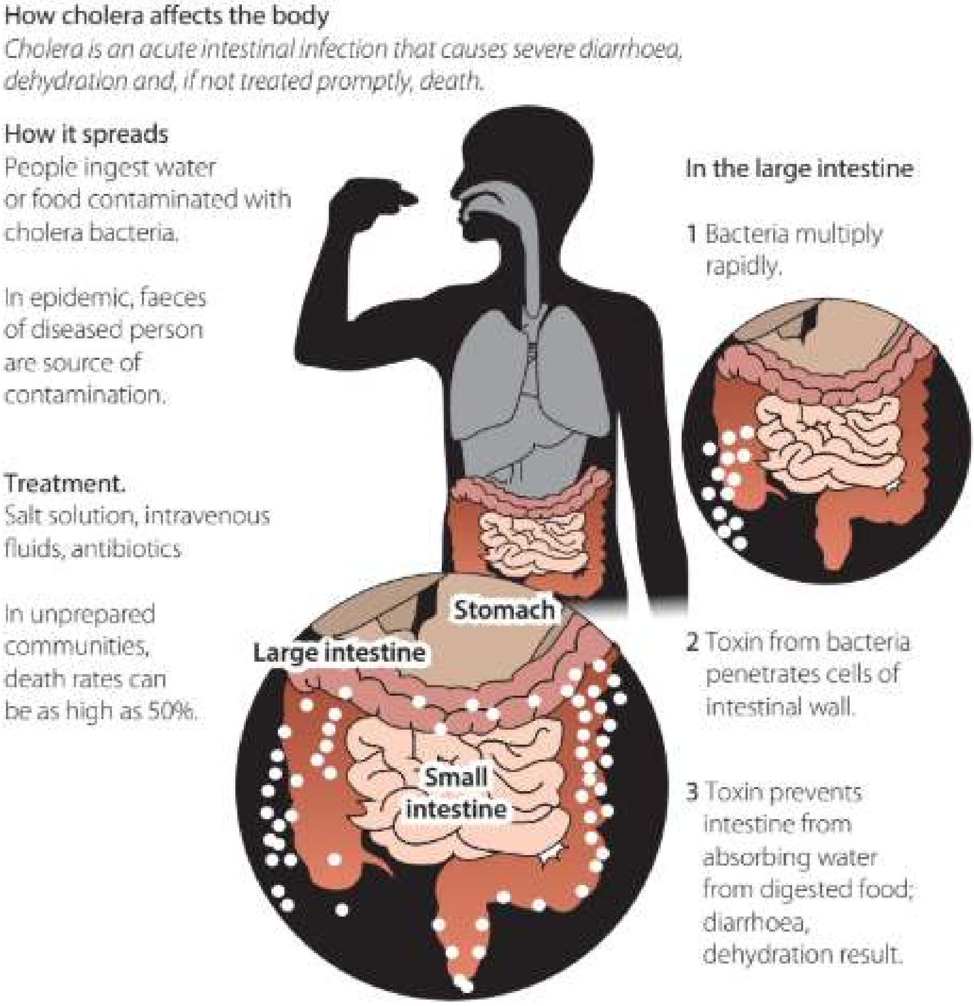


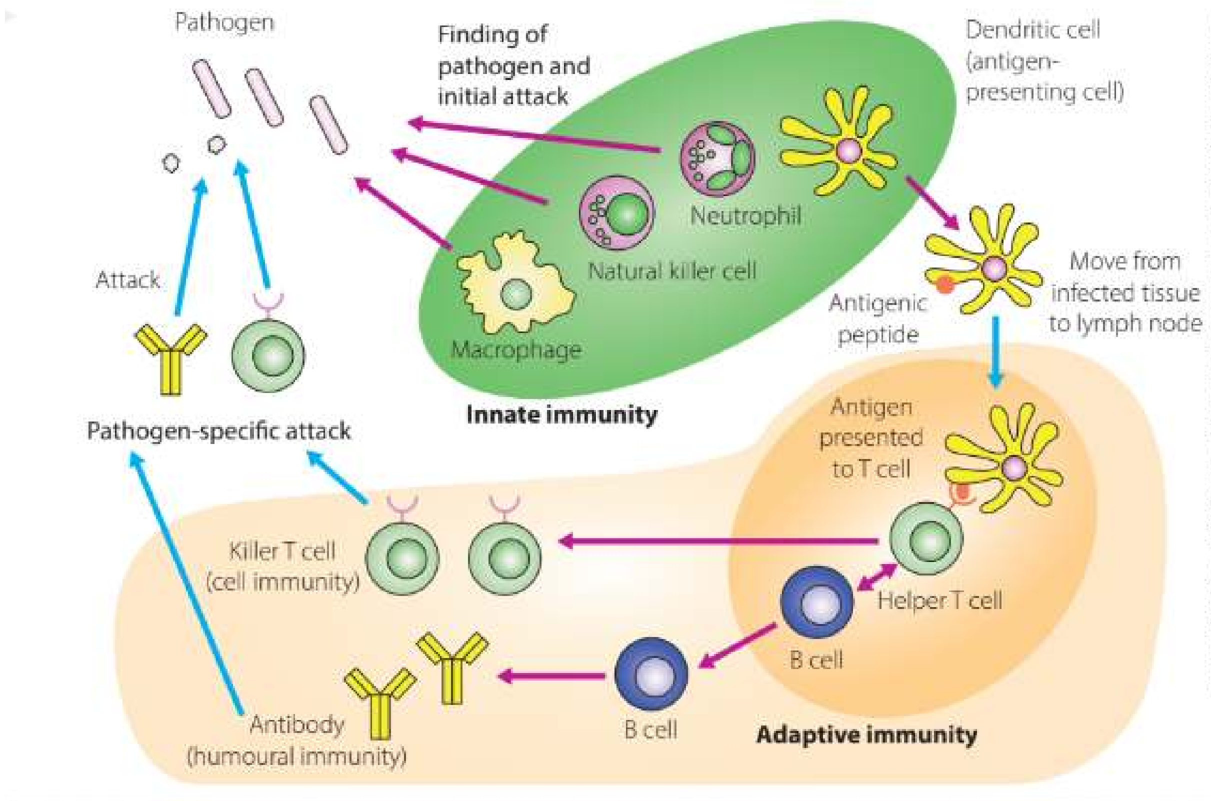
FIGURE 12.27 The pathogenic effects of cholera bacteria on the human gut

- Microbiome of gut - major role in regulating immune response to diseases
- All humans provide microhabitats for many microorganisms
- Adaptive immune response
  - Does not reach a level fast enough to allow person to recover before diarrhoea and vomiting kills them
  - Bodys microbiota also outcompeted in the process

Modelling the Innate and Adaptive Immune Systems:

- Using:
  - Diagrams
  - Analogies

Diagram:

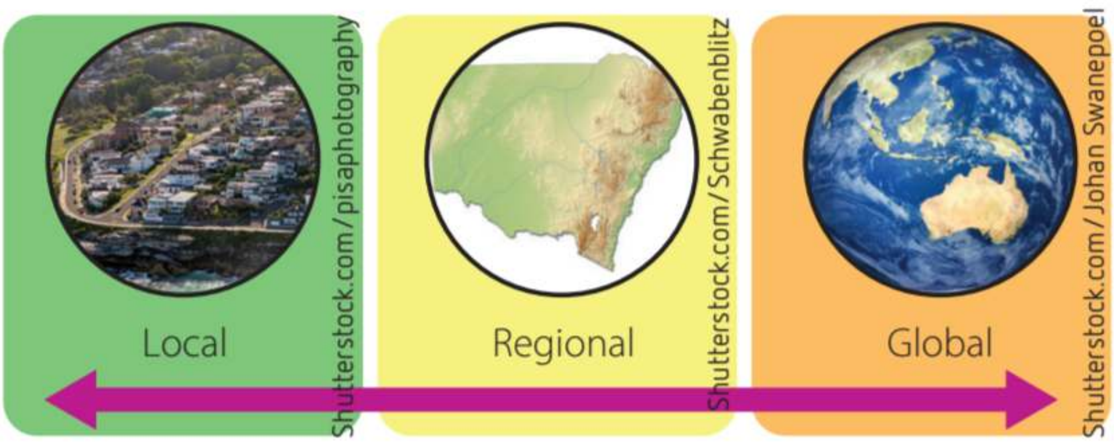
- 

Analogy:

- Analogy in one of dot point booklets for mod 7

Limiting the Spread of Infectious Disease

Factors involved in Monitoring and Control:

- 

Local Factors:

- Affect local areas including:
  - Neighbourhoods

- Village
- Town
- City
- Major Factors:
  - Sanitation:
    - Especially waste and sewage disposal
  - Overcrowding:
    - Increases chance of host to host transmission
    - Poor communication on networks, roads:
      - Limit access to:
        - medical treatment
        - Hospitals
        - Medical information
  - Animal husbandry:
    - Increase transmission of:
      - Avian swine flu to humans
  - Cultural, spiritual beliefs:
    - Influence attitudes towards western medical practices

Regional Factors:

- 5 regions:
  - Africa
  - The americas
  - Asia
  - Europe
  - Oceania
- Regions characterised by geographical features:
  - Mountains
  - Deserts
  - Rainforests
  - Grasslands
- Determine if a population in region is:
  - highly mobile (moving)
  - Isolated
- E.g.:
  - People in mountainous regions, on islands
    - Have isolation factor

    - Reduces their exposure to infected people
- Local seasonal variations in:
  - Temperature
  - Precipitation

  May influence vector availability

Global Factors:

- Increased movement of people around world:
  - Harder to control spread of disease
  - Easier to transmit disease
- Misuse of antibiotics, other antimicrobial medications:
  - Increase in resistant bacteria
- Easy communication through internet:
  - May provide accurate up to date data of disease outbreaks as they occur
  - May provide misinformation of disease spread, causes, measures

Disease Transmission Factors:

Pathogen Features:

- Growth:
  - How quickly and easily pathogen replicates
- Survival:
  - Limits:
    - how for pathogen travels
    - Pathogen opportunity to infect
  - Adaptations of pathogens:
    - In beginning of mod
  - Determines method of transmission:
    - Pathogen ❌ survive outside host:
      - Direct transmission
    - Pathogen ✅ survive outside host:
      - Indirect transmission
- Persistence:
  - Allows them to multiply over long period of time
  - Evade host immune system:
    - E.g: dormant cells with slowed metabolic processes - lying low
      - Effects on people:
        - Limited damage to host

- Infected host acts as reservoir for infection

- Virulence factors:
  - Strategies used by pathogens to cause disease
- Staying alive in places:
  - Forming reservoirs in:
    - Food
    - Water
    - The environment

Pathogen Method of Transmission:

- Method of transmission:
  - Affects spread of disease
- Diseases travel further by indirect transmission
- Being transferred directly from hosts to host:
  - Does so if not environmentally resilient

Host Factors:

- Concurrent illness
  - Illness happening at same time as new illness
  - May reduce effectiveness of host's immune system
- Malnutrition:
  - Decreases hosts ability to avoid disease
- Pharmaceuticals:
  - May lower body's defence against pathogens

Environmental and Geographic Factors:

- Pathogen builds up large reservoir in environment:
  - Greater risk of outbreak
  - E.g.:
    - Natural disasters → poor sanitation
      - Increase in cholera causing bacteria in local waterways
- Environmental conditions may favour pathogen survival in environment:
- E.g.:
  - Mosquito need warm weather + water body
    - To produce and transmit malaria parasite

Societal + Population Factors:

- Population density:

    - How physically close people are living to one another
    - Diseases spread quicker when:
        - People live closely together
            - Contact between individuals → more likely → increased chance of exposure to disease
- High population density:
    - Influenced by:
        - Overcrowding
        - Lack of infrastructure
            - E.g. poor sanitation
- Movement of people:
    - Infected people travel → spread of disease to contacted people
    - People exposed to disease which have never encountered disease:
        - Have no immunity to it
- Proportion of population immune to disease:
    - Herd immunity:
        - If most of population is immune to disease
        - Any susceptible people are prevented from contacting with pathogen
        - Happens because:
            - Immune people break up transmission of pathogen
        - Amount of people to be immune:
            - Dependent on:
                - Individual pathogen
                - How infectious pathogen is
- Lack of education:
    - On vaccines
    - Factors in epidemic

    Leads to poor decision making by public
- Socioeconomic factors:
    - Poverty
    - Lack of access to:
        - Vaccines
        - Education
- Lack of genetic diversity:
    - Communities with a lack of variation in innate
    - And no adaptive immunity

    Are at greatest risk of disease outbreaks

Preventing Spread of Disease:

Hygiene:

Personal:

- Hands washed with soap and water:
    - Before:
        - Eating food
        - Preparing food
    - Prevents spread of pathogens that
        - Cause symptoms like diarrhoea
- Body + hair + teeth washed, cleaned regularly:
    - Prevents build up of a lot of pathogens (mostly bacteria) to cause disease
- Always cough + sneeze into handkerchief or tissue:
    - Prevents airborne droplets from spreading to others
    - No sneeze into hands as
        - Will transmit pathogens to whatever touched

Community:

- Proper sewage + garbage disposal:
    - Reduce risk of pathogen numbers:
        - increasing, spreading
    - May include use of UV disinfection for:
        - Management of pathogens in wastewater
- Sterilisation + disinfection of equipment in:
    - Hospitals
    - Doctors surgeries
    - Dentists
    - Hairdressers, barbers
- City planning:
    - Reduces overcrowding:
        - Reduces risk of disease transmission

Clean Food and Water:

- Guidelines to be followed by food handlers:
    - For:
        - Storing
        - Preparing

        - Serving food
    - To:
        - Control transmission of disease from food sources
- Water quality maintained
    - To reduce risk of:
        - pathogens reproducing
        - Transmission of pathogens in contaminated water
    - Through strict standards and guidelines

Quarantine:

- Isolation of individuals who have infectious disease
    - from others
        - for a period of time
- Ensures individuals are free from disease before contacting with communities
- Minimises risk of exotic pests and diseases entering a region
- Only last for the duration where:
    - Disease is infectious to communities

Small-scale:

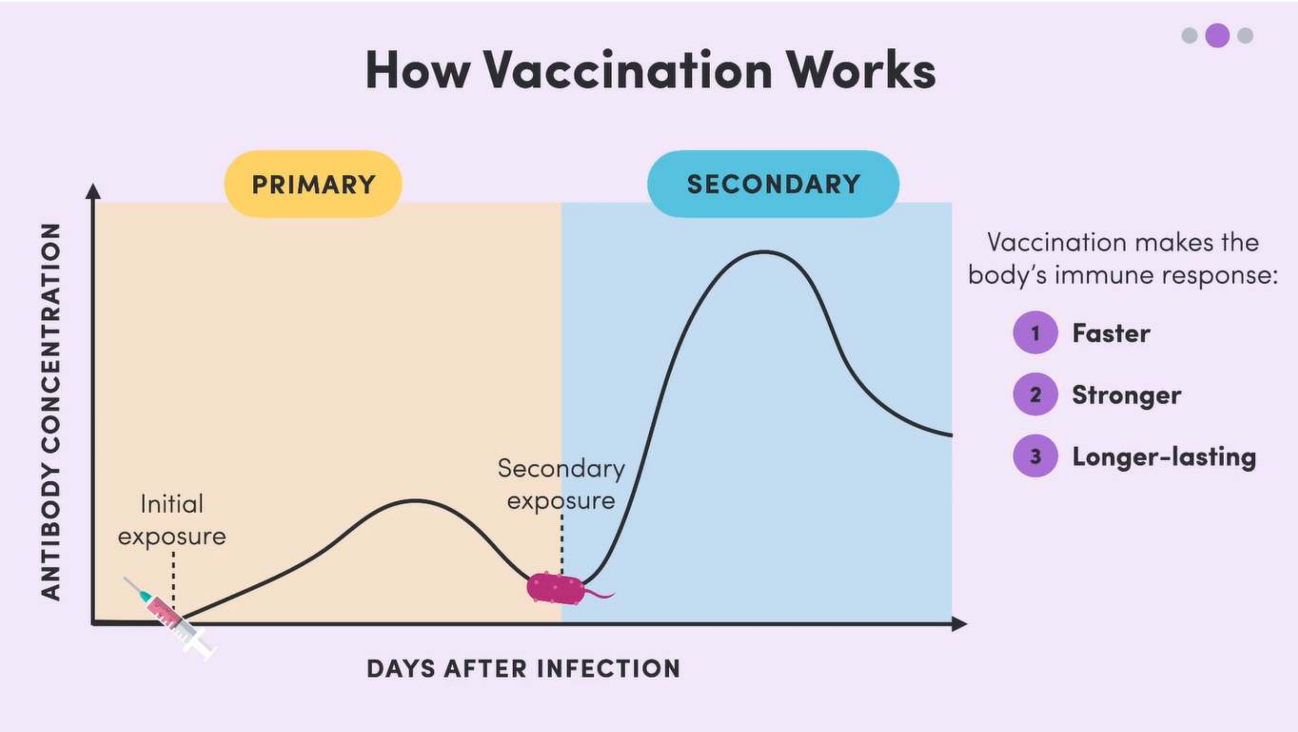
- Confining infected patients:
    - To their homes
    - To their hospital bed

Large-scale:

- Confining whole communities:
    - Schools + workplaces closed
    - Borders cut and patrolled with:
        - Strict exit + entry criteria based on health
- Goods and transport:
    - Banning the movement of fruit across borders to:
        - Protect against fruit flies

Vaccination:

- Involves the introduction of a vaccine into the body
- Immunisation:
    - Process in which body reacts to vaccine by going through immune response
        - Response produces memory cells for antigen
        - ∴ immunity to body

- If antigen enters body again:
    - Secondary immune response:
        - avoid worst symptoms of disease
        - Destroy pathogen before experiencing symptoms of disease
- Immunity produced by vaccines if lifelong
- 

Vaccines:

- Have cultures of microbes which are:
    - Living but weakened:
        - ∴ harmless
        - E.g. rabies, poliomyelitis, measles
    - Dead:
        - E.g. typhoid, whooping cough
- May have modified toxins (toxoids)
    - For: tetanus, diphtheria
    - May be given:
        - Orally
        - Injection
        - Scratching skin surface
- Have antigens of disease
    - Cause body to undergo immune response
        - Produce memory cells for pathogen antigen
- Booster vaccines:

- Given over a number of years
- Makes vaccination more effective
- More memory cells produced as immune system recognises antigen more rapidly
    - Able to provide immunity for long time
- Immunity of body to disease decreases over time:
    - Booster vaccines given to
        - Increase amount of circulating memory cells

ActIve Acquired Immunity:

- When:
    - immune response occurs
    - Memory cells produced
- Lasts long term
    - As long living memory cells made
- Naturally induced:
    - Through direct exposure to pathogen
    - Patient suffers symptoms of disease
- Artificially induced:
    - Through use of vaccines
    - Patient does not suffer symptoms of disease

Passive Acquired Immunity:

- Introduction of antibodies in body
- Antibodies produced by other organism who had disease
- Immunity lasts short time as
    - No memory cells produced by body itself

Public health Campaigns:

- Different health campaigns used for diff communities as
    - They may be affected by different diseases
- Campaigns vary if they are
    - short term immediate response to infection
    - Long term measure
- E.g. Vector borne disease campaigns
    - Malaria (mosquito)
        - Measures:
            - Fumigating homes
                - Disinfecting or purifying homes with fumes of certain chemicals

    - Cleaning out stagnant water
        - Water in confined space having no current or flow
- RICE:
    - Resolution:
        - of governments and health organisations (WHO)
            - To find solutions to infectious disease
    - Information:
        - In form of epidemiological studies + scientific studies of:
            - pathogen
            - Its mode of transmission
        - Done so solution is based on accurate data
            - Includes:
                - National funding for scientific research
                - + training of scientists, medical personnel
    - Coordination:
        - Of efforts on local, regional and global scale so
            - Resources used efficiently
    - Education:
        - Of human population on local, regional, global scale
            - For factors affecting disease transmission
- Government regulations:
    - Standardised practices for:
        - Handling
        - Cooking
        - Storing

        Food
    - Medical settings (hospitals, surgeries, clinics):
        - Sterilising equipment
        - When health workers move from patient to patient
    - Waste disposal
    - Treating drinking water
    - Sewage removed + treated
    - Reporting diseases to authorities if detected
        - Allows for:
            - early detection of diseases
            - Appropriate measures to be put in place
        - E..g. of diseases:

        - Measles, botulism, Cholera, meningococcal infection, whooping cough, malaria
- Public immunisation programs:
    - Government funded + enforced
    - Childhood immunisation
        - For:
            - Diphtheria, tetanus, whooping cough, measles, mumps, rubella

Pesticides:

- Chemicals used to kill pests of plants + animals
    - Including:
        - Pathogens
        - vectors
- 3 groups:
    - Insecticides:
        - Kill insects
    - Fungicides:
        - Kill fungus pathogens
    - Herbicides:
        - Kill weeds
- Benefits:
    - Protects against:
        - Vector borne diseases:
    - Protects
        - Crop
        - Livestock
        - Defending health of communities by sustaining food source
- Problems:
    - vectors or weeds developing resistance to pesticide
    - Damaging to environment

Genetic Engineering:

- Altering genetic composition of organism
- Can make organisms resistant to diseases
- Disable vectors from spreading disease
- Modify animals to:
    - Produce biomed products
- Prevents:

    - Disease occurring
    - Prevents + controls spread of disease
- Transgenic species:
    - Organisms with genes from other organisms put into their own genes
- Problems:
    - Genetically altered organisms developing resistance to pesticides
    - Effect of organisms on environment + biodiversity
    - Ethical issues:
        - Public labelling of products derived from genetically modified organisms

Pharmaceuticals for Controlling Disease:

Antivirals:

- Used to control viral infections
- Don't kill viruses
- prevent virus development inside infected cells
- Don't cure disease
- Only slows down virus progress
    - Allows for body’s natural defences to take over
- If taken early
    - Symptoms are milder, of shorter duration
- Stop spread of viral diseases
- Problems:
    - Access to antivirals
    - Compliance with dosing
    - Viruses developing resistance
- Antivirals most commonly used for:
    - HIV (virus) causing AIDS (disease)
    - Seasonal influenza A
    - Herpes
    - Hepatitis B and C
- Why it's hard to develop drugs to stop virus reproduction:
    - Viruses replicate using host cell
    - Hard to stop replication without killing host cells
- Antivirals may target different stages of virus replication cycle

How Antivirals may work:

- Prevent viruses from entering host cells by:

- binding to receptors
- Stops virus from getting a hold on cell
- Inhibit enzymes that:
- catalyse reproduction of viral genome
- Blocking:
  - Transcription
  - Translation

    Of viral proteins
  - As they dont have ribosomes:
    - They need ribosomes of host cell to make their protein coat
- Preventing virus from leaving cell:
  - Stopping spread of virus to other cells
- Grabbing virus while outside cell
  - By targeting capsid proteins (on external protein coat)

Antibiotics:

- Drugs used to control bacteria infections
- Usually are extracted compounds made by fungus or bacteria
- Work by:
  - Killing
  - Slowing down growth of bacteria
    - To be killed by immune system
- To work, they have to:
  - Not damage host organism
  - Specifically target:
    - Bacterial features:
      - Cell wall synthesis
      - Nucleic acid synthesis
      - Protein synthesis
      - Certain metabolic pathways
- Most antibiotics:
  - Used to treat most bacteria as
    - Most bacteria share same bacterial features
- Not effective against viruses as they are made for bacteria
- Most effective when:
  - Used only for bacterial infections
  - Used for killing bacteria instead of preventing growth

- Narrow-spectrum used to target the specific pathogen
- They can go to site of infection to kill bacteria
    - Barriers: blood-brain, necrotic, granulating tissue
- Therapeutic blood levels maintained
- Whole course dosage is taken:
    - Reduces risk of bacterial resistance

How Antibiotics work - Penicillin:

- Prevents bacterial cell walls from developing properly
- Inhibits synthesis of Peptidoglycan
    - Mesh-like substance made up of:
        - Sugars
        - Amino acids
    - Ensures bacterial cell has water balance
- Penicillin weakens cell wall of bacteria
    - Water goes into bacteria
    - Cell then pops and dies
- Does not affect host of bacteria, only affects bacterial cell
    - As animals don't produce peptidoglycan

Antibiotic Resistance:

- Bacteria can become resistant to antibiotics
    - By overuse of antibiotics
- Happens when:
    - Failing to complete antibiotic course
        - Allows resistant bacteria to survive + replicate
    - Spontaneous Mutation or gene which provides resistance to antimicrobial substance
        - Gets selected and passed onto reproduced bacteria
- Bacterial Mutations:
    - Deactivating, degrading enzymes made by antibiotics
    - Efflux pump to push antibiotics out
    - Entry blockers
        - Preventing antibiotic entry

Environmental Management + Quarantine Methods:

Ebola Virus Disease 2014-2016:

- Ebola virus → ebola virus disease

- Is ssRNA rivus:
    - Single stranded + RNA
- Spread when:
    - People have direct contact with infected
        - Body fluids
        - Mucous membranes
        - Items contaminated by above
        - Sexual contact
- First appeared in africa 1976
- Large outbreak in 2014-2016
- Transmission:
    - Most likely to be through close human contact with wild animals
        - Who are reservoirs of virus
    - E.g. consuming their meat
    - Can be through:
        - Rituals involving direct contact with body of affected person
- Reservoirs:
    - Body tissues + fluid of apes
    - Antelope
    - Fruit bats - suspected to be natural reservoir
    - Porcupines
- Symptoms:
    - Start as:
        - Fever
        - Tiredness
        - Headache
        - Sore throat
    - Progress to:
        - Vomiting
        - Rash diarrhoea
        - Oozing of blood from mucous membranes + faeces
            - Bleeding from:
                - Clotting failure from liver damage
                - (healthy liver makes clotting factors)
- Management of disease:
    - Supportive with
        - Intravenous, oral fluid to:

            - replace lost blood
            - Maintain circulating blood volume
        - Antibiotics - manage secondary bacterial infections
        - Antipyretics - fever
        - Analgesia - pain

Control of Epidemic:

Environmental Management:

- Controls:
    - Preventing transmission of pathogens in clinical setting of epidemic
        - Using procedures + protocols
- Administrative controls:
    - Initial organisation of response
    - Deciding which people responsible for:
        - what parts of main management strategy
        - Getting funding
- Environmental + engineering controls:
    - Overseeing of facilities for:
        - barrier nursing
            - (giving care in strict infection control conditions)
        - Workspaces
- Water + hygiene controls
- Hand hygiene
- Safe waste management
- Overseeing + maintenance of:
    - Personal protective equipment (ppe)
- Puncture-proof containers:
    - For disposal of all sharps
    - Preventing needle-stick injuries
- Contaminated environmental surfaces:
    - Disinfected with 0.5% chlorine solution asap after patient exposure
- All surfaces cleaned with detergent one a day
- Cleaners wore PPE
- When Lab staff working with equipment which could disperse virus through air
    - They wore respirators
- Samples from infected humans + animals
    - Only handled by trained staff

    - Processed in properly equipped labs
- Tissues + body fluids for disposal
    - Placed in clearly marked bags for incineration (destroying through burning)
- Counselling for:
    - Ebola survivors + sexual partners to:
        - Use condoms until 2 negative results obtained for virus presence in semen
- Long-term measures:
    - Reducing wildlife-human transmission by:
        - Making sure meat cooked thoroughly
        - Wearing gloves when handling animal reservoirs
- Managing regions cultural + spiritual beliefs:
    - Working with community leaders
        - For managing problem of mistrust in government + foreign workers
- Reducing human-human transmission:
    - Through good hygiene + environmental control procedures
        - During + after infection
    - Safe sex and hygiene:
        - For 12 months after infection
- Outbreak containment:
    - Prompt + safe burial of dead

Quarantine Methods:

- Isolating patients in a single room or
- Providing at least 3 metres of space between patient beds
- Same:
    - clinical staff assigned to a single patient
    - Medical equipment assigned to a single patient
- All visits restricted
    - Excluding parents of a sick child
- Exposed people:
    - Isolated + monitored for 21 days after exposure
    - Help reduce transmission
- Border checkpoints:
    - With armed guards
    - To prevent movement of people in and out of quarantine area

Incidence + Prevalence of a Disease:

- Incidence + prevalence of disease:

    - Determines pool of pathogens already present in population
- Mobility:
    - Determines how easy it is for pathogen to spread
- % of population immunised against infectious disease:
    - Determines what proportion of population vulnerable + may act as carriers for pathogen

Incidence:

- Number of new cases of disease occurring during a specified time
- Can be thought of as infection rate
    - Or probability (risk) of contracting disease
- Can be expressed as %
    - or as number per 100 000 (or 1000 or 10 000 …) of population
- To calculate incidence of disease as percentage:
    - $\frac{number\ of\ new\ cases\ during\ specified\ time}{size\ of\ population\ at\ start\ of\ monitoring\ period} \times 100$

Prevalence:

- Refers to proportion of population that have disease at a particular point in time
- Refers to all cases
    - Previous cases + current
- Expressed as %
- To calculate prevalence of disease as percentage:
    - $\frac{all\ new + previous\ cases\ during\ a\ time\ period}{population\ during\ the\ time\ period} \times 100$

Mobility:

- How mobile a population is
- How much a population moves
- Rate of immunisation:
    - At what rate a population is being immunised against a disease
- Herd immunity:
    - Occurs when a significant proportion of a population is immunised against disease
    - Relies on high numbers of people being vaccinated
        - To reduce chances of unvaccinated people coming into contact with pathogen

Mosquito Borne Diseases:

Examples:

- Malaria:

- Caused by:
    - Plasmodium Falciparum
- Vector:
    - Anopheles mosquito
- Dengue Fever:
  - Caused by:
    - Flavirus
  - Vector:
    - Aedes mosquito

Factors Affecting their Transmission:

Environmental:

- Temp
- Excess Rainfall:
  - Leads to creation of natural water pools + puddles
    - Serve as breeding sites for mosquitoes
  - Increase transmission of mosquito borne disease
- Humidity
- Influence mosquito rates of:
  - Development
  - Breeding
  - Mortality
- Typically affect tropics as climate favourable
- Mosquito-borne diseases → tropical diseases
- Climate change + global warming:
  - Temps rise:
    - Mosquito territories increase
    - Distribution of mosquito borne disease increase
  - Climate in places where mosquito borne disease prevalent:
    - Will cause longer breeding seasons
    - Increased hatch rates of mosquitoes
    - ∴ higher rates of mosquito borne disease

Social:

- Most affected areas:
  - 3rd world countries
    - Unable to provide adequate:

- health care
    - eradication methods
    - Education programs
- Presence of:
  - Containers of starred water
  - Discarded junk
  - Dams

  Increase breeding grounds of mosquitoes
- Unplanned urbanisation
  - Where normally lack of waste removal infrastructure
- Globalisation:
  - Increased movement of people over large areas in shorter times
  - Increases distribution of mosquito-borne disease
    - As people who have never been exposed may be exposed to disease
    - They don't have any immunity to them

Predicting + Controlling Spread of Disease:

Historic Control:

- First quarantine measures:
  - Used in dubrovnik, croatia in 1377
  - to control plague outbreak
- First permanent plague hospital"
  - Lazaretto
  - Open don island of santa maria di nazareth in 1423
  - Was strategically located so they would have natural barrier like sea, river, or mountain
    - To separate them from populated areas

Cultural Control:

- Philippines:
  - Traditional foods → large part in preventing + curing disease
  - Garlic + onions
    - Believed to lower blood pressure
  - Is now known that onions have Quercetin:
    - Natural supplement
      - Lowers blood pressure
- Egypt:
  - Queen cleopatra:

        - Used mosquito nets
        - Even when no one knew about malaria + mosquitoes

Aboriginal Protocols in development of Medicines:

Bush Medicine:

- Bioactive compounds in bush medicines
- Alkaloid compounds from Moreton Bay Chestnut or black bean
    - Showing promise in management of HIV/AIDs
- Indigenous people use specialised preparation techniques:
    - to minimise harm effects of bush medicine

Smoke Bush:

- Some species have:
    - Big + wooly flowers that resemble drifting smoke
- Used by aboriginal people for healing
- Properties discovered by scientists
    - Potential uses against cancer + HIV/AIDS

Importance of Intellectual Property:

- To protect traditional knowledge and cultural material
- Prior consent minimises cultural harm + offence
- Ensures intellectual property is rightfully attributed
- Ensures it is not misused

Antibiotics are drugs made from extracted compounds of fungi or bacteria, that are used to treat and control bacterial infections. These drugs are only effective against bacteria and are the most effective when they are used to kill bacteria instead of preventing its growth within the host individual. An example of an antibiotic includes penicillin, which is derived from the extract of a fungus and works to stop the synthesis of Peptidoglycan, a compound used to maintain the water balance within bacteria. This allows water to go into the bacterial cell and eventually pop, killing the cell. Antivirals however, are drugs used to control infections caused by viruses. They only work against viruses and are used to control and prevent the development of viruses inside host cells, not being able to kill them, contrasting with the use of antibiotics. An example of an antiviral includes those which are used to control the spread of the HIV virus, preventing it from entering by binding to receptors, stopping the virus from getting a hold on the cell and invading it.

Antibiotic treatment has a limited scope in targeting bacterial cells that are present within host cells, as specifically targeting only the bacterial cell without inflicting damage to the host cell is difficult. As bacteriophages are viruses that use bacteria as their own host cell, they may be used to target and kill bacterial cells residing in an individual's body cells, not normally able to be targeted using antibiotics. Antibiotic resistance is also a prevalent issue arising due to the overuse of antibiotics, failure to complete full doses and random mutations occurring within the bacteria, leading to them evolving and becoming resistant to the pharmaceutical. The use of bacteriophages directly counters this issue as when the bacteria itself evolves, the bacteriophage that kills the bacteria also evolves, still being able to damage bacteria and treat associated bacterial infections.